Instituto Superior Técnico

Telecommunications

Lecture notes

Signals, digitization, modulation, noise,
digital detection and system operation

Based on the 2025–2026 lectures and problem sheets
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Student edition
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How to use these notes

These notes develop the material in the seven weekly Telecommunications lectures, the supplementary signals-and-systems deck, and the thirteen corrected problem sheets supplied with the course. They are intended for students who have already encountered calculus, complex numbers, elementary probability and linear systems. The mathematical tools are recalled where they are used.

Read a chapter first for the physical question being answered, and then reproduce the derivations with pen and paper. In a worked example, stop before the solution and identify the relevant model, units and assumptions. Finally, attempt the corresponding problem sheet. The notes explain the methods and work through representative problems; they do not replace the original sheets as a source of independent practice.

The structure follows the lecture sequence, with longer topics divided into manageable chapters. In particular, pulse shaping belongs naturally with digitization and line coding, while detection is developed after the noise model has been established. The additional explanations in the recovered digital-transmission deck are incorporated into the discussion of receivers and signal space.

Conventions matter. The same letter can mean different things in different slides. Here, frequencies are in hertz, the sinc is normalized, and $N_0/2$ always denotes the two-sided PSD of real white noise. Message bandwidth, channel bandwidth, bit rate and symbol rate are kept distinct. Appendix C explains substantive corrections and ambiguities in the supplied material, including those that change numerical answers.

What the examples assume. Unless stated otherwise, filters and synchronization are ideal, symbols are equiprobable, and electrical energies are normalized to a $1\,\Omega$ load. Approximate formulas are identified as such. A calculated minimum at the Nyquist rate is an ideal limiting value: a practical design needs filter transition bands and implementation margin.

Lecture material	Chapters in these notes	Problem sheets
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Week 6, both versions	9–10: matched filters, signal space, digital modulation	10–11, 13
Week 7	11: operation, capacity and error control	Integrated exercises

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Notation and units

Symbol	Meaning
$x(t), X(f)$	Signal and Fourier transform; capital $X(t)$ is instead a random process when explicitly introduced.
$m(t), P_m, B_m$	Message, its mean-square power, and its one-sided baseband bandwidth.
B_{BB}	Baseband channel bandwidth: support extends from $-B_{\mathrm{BB}}$ to $+B_{\mathrm{BB}}$.
B_T	Width of a positive-frequency passband, e.g. from $f_c - B_T/2$ to $f_c + B_T/2$.
f_a, T_a	Analog sampling frequency and sampling interval, $T_a = 1/f_a$.
R_s, T_s	Symbol rate (baud) and symbol interval, $T_s = 1/R_s$.
R_b, T_b	Transmitted bit rate and bit interval. For uncoded M-ary signaling, $R_b = R_s \log_2 M$.
L_q, ν, Δ	Number of quantization levels, bits per sample, and uniform quantizer step; $L_q = 2^\nu$.
M	Number of modulation symbols. This need not equal L_q.
E_p, E_s, E_b	Pulse energy, average symbol energy, and average energy per transmitted bit.
$S_x(f), \Psi_x(f)$	Power spectral density and energy spectral density.
$N_0/2$	Two-sided real AWGN PSD. N_0 has units W/Hz, equivalently J.
α	Raised-cosine roll-off factor, $0 \le \alpha \le 1$.
k_f, k_p	FM sensitivity in Hz per message unit; PM sensitivity in rad per message unit.
$\Delta f, \beta$	Peak frequency deviation and, for a tone, FM index $\beta = \Delta f / f_m$.
γ	Analog reference SNR $P_R/(N_0 B_m)$; generally different from receiver-input SNR.
P_b, P_s	Bit and symbol error probabilities.
r_c	Channel code rate: information bits per coded bit.

For a voltage signal across a resistance R, physical power is $\langle |x|^2 \rangle / R$ and energy is $\int |x|^2 \, \mathrm{d}t / R$. With normalized resistance, these factors are suppressed. An amplitude in a unit-energy signal basis has units $\sqrt{\mathrm{J}}$; its squared magnitude is an energy, not an instantaneous voltage.

Chapter 1

Signals and Fourier analysis

Course connection: Week 1, supplementary deck, Sheets 1–2..

1.1 The communication problem

A communication system conveys a message through a physical medium. The transmitter converts that message into a waveform suitable for the medium. The channel attenuates and distorts the waveform and introduces noise. The receiver uses the available observation to estimate either the original waveform or the transmitted symbols.

An analog message may vary continuously in amplitude and time. A digital message belongs to a discrete alphabet. The electrical waveform carrying digital information is still a continuous-time physical signal: “digital” describes the information representation and decision process. Regeneration can remove accumulated amplitude perturbations when the decisions are correct. A wrong decision becomes a symbol error and cannot be repaired merely by amplifying the output.

A useful link description specifies the message bandwidth, allowed transmission power, channel response, noise statistics and performance target. For analog systems the target is often output SNR or distortion. For digital systems it is usually a bit error probability at a stated information rate.

1.2 Elementary signals

We use

$$\operatorname{rect}(u) = \begin{cases} 1, & |u| < 1/2, \\ 0, & |u| > 1/2, \end{cases} \qquad \operatorname{tri}(u) = \max(1 - |u|, 0), \tag{1.1}$$

$$\operatorname{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \qquad \operatorname{sinc}(0) = 1. \tag{1.2}$$

Values at isolated pulse endpoints do not affect integrals. The unit step $u(t)$ is zero for $t < 0$ and one for $t > 0$. The Dirac impulse is a distribution specified by

$$\int_{-\infty}^{\infty} x(t)\delta(t - t_0)\,\mathrm{d}t = x(t_0). \tag{1.3}$$

It has unit area, not unit height. It idealizes a pulse whose duration is short compared with the time scales being studied.

1.3 Energy, power and RMS amplitude

The signal energy and time-average power are

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, \mathrm{d}t, \tag{1.4}$$

$$P_x = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \, \mathrm{d}t. \tag{1.5}$$

An energy signal has $0 < E_x < \infty$ and therefore zero time-average power. A power signal has $0 < P_x < \infty$ and infinite energy. Some idealized signals fit neither class. For a periodic signal of period T_0,

$$P_x = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} |x(t)|^2 \, \mathrm{d}t, \qquad x_{\mathrm{rms}} = \sqrt{P_x}. \tag{1.6}$$

A sinusoid $A\cos(2\pi f_0 t + \phi)$ has power $A^2/2$, whereas a rectangular pulse of height A and duration T_p has energy $A^2 T_p$. If $x(t) = m_x + \widetilde{x}(t)$ and $\langle \widetilde{x} \rangle = 0$, then $P_x = |m_x|^2 + P_{\widetilde{x}}$. This DC/AC decomposition later explains carrier power and spectral lines.

1.4 Fourier transform and interpretation

Our transform pair is

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} \, \mathrm{d}t, \qquad x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} \, \mathrm{d}f. \tag{1.7}$$

$X(f)$ gives the complex weight of each frequency. Its magnitude describes the relative strength of a component and its phase describes timing relative to the chosen origin. A real signal obeys $X(-f) = X^*(f)$, so magnitude is even and phase is odd wherever defined.

Absolutely integrable signals have an ordinary Fourier transform. Finite-energy signals admit an L^2 transform, while periodic signals and impulses require the distributional interpretation. One need not force all three cases into the same ordinary integral.

Time domain	Frequency domain
$\delta(t)$	1
1	$\delta(f)$
$\mathrm{rect}(t/T)$	$T\,\mathrm{sinc}(fT)$
$\mathrm{tri}(t/T)$	$T\,\mathrm{sinc}^2(fT)$
$e^{-at}u(t)$, $a > 0$	$1/(a + j2\pi f)$
$\cos(2\pi f_c t)$	$[\delta(f - f_c) + \delta(f + f_c)]/2$
$\sin(2\pi f_c t)$	$[\delta(f - f_c) - \delta(f + f_c)]/(2j)$

Worked example 1.1 (A decaying pulse; Sheet 1, Problem 1.8)**.**
Find the transform and energy of $x(t) = e^{-at}u(t)$, $a > 0$.

Solution. The step restricts the integration interval:

$$X(f) = \int_0^{\infty} e^{-(a+j2\pi f)t} \, \mathrm{d}t = \frac{1}{a + j2\pi f}, \qquad E_x = \int_0^{\infty} e^{-2at} \, \mathrm{d}t = \frac{1}{2a}.$$

Thus $|X(f)|^2 = 1/[a^2 + (2\pi f)^2]$. Increasing a shortens the pulse and broadens its spectrum. Its amplitude spectrum falls to $1/\sqrt{2}$ of its DC value at $f = a/(2\pi)$.

1.5 Transform properties as a calculation method

If $x(t) \leftrightarrow X(f)$ and $y(t) \leftrightarrow Y(f)$, then

$$x(t - t_0) \leftrightarrow e^{-j2\pi f t_0} X(f), \tag{1.8}$$

$$x(at) \leftrightarrow \frac{1}{|a|} X(f/a), \quad a \neq 0, \tag{1.9}$$

$$x(t) e^{j2\pi f_c t} \leftrightarrow X(f - f_c), \tag{1.10}$$

$$x(t) \cos(2\pi f_c t) \leftrightarrow \tfrac{1}{2}[X(f - f_c) + X(f + f_c)], \tag{1.11}$$

$$\frac{\mathrm{d}x}{\mathrm{d}t} \leftrightarrow j2\pi f X(f), \tag{1.12}$$

$$(x * y)(t) \leftrightarrow X(f) Y(f), \tag{1.13}$$

$$x(t) y(t) \leftrightarrow (X * Y)(f). \tag{1.14}$$

Linearity allows these rules to be combined. The duality rule is $X(t) \leftrightarrow x(-f)$ when both sides are interpreted consistently. Integration from $-\infty$ corresponds to multiplication by the transform of $u(t)$:

$$\int_{-\infty}^{t} x(v)\,\mathrm{d}v \;\leftrightarrow\; \frac{X(f)}{j2\pi f} + \frac{X(0)}{2}\delta(f),$$

where the division is understood in the principal-value sense. The DC term matters when $X(0) \neq 0$.

Key idea. Before evaluating a new integral, ask whether the waveform is a shifted, stretched, modulated or differentiated version of a known one.

1.6 Fourier series and Parseval's relations

A periodic signal has the representation

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_0 t}, \qquad c_n = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-j2\pi n f_0 t}\,\mathrm{d}t. \tag{1.15}$$

Its transform is $X(f) = \sum_n c_n \delta(f - n f_0)$. A periodic pulse train with pulse height A, width τ, period T_0 and pulses centered at multiples of T_0 has $c_n = A(\tau/T_0)\,\mathrm{sinc}(n\tau/T_0)$.

The energy and power versions of Parseval's identity are

$$E_x = \int_{-\infty}^{\infty} |X(f)|^2\,\mathrm{d}f, \qquad P_x = \sum_{n=-\infty}^{\infty} |c_n|^2. \tag{1.16}$$

The second expression uses Fourier-series coefficients. Squaring a sum of Dirac impulses is not a valid way to derive a periodic signal's power.

1.7 What does bandwidth mean?

A strictly bandlimited baseband signal has $X(f) = 0$ for $|f| > B$. A time-limited rectangular pulse is not strictly bandlimited. Its first positive spectral zero is $1/T_p$, so one may quote the approximate baseband first-null bandwidth $1/T_p$, or the full main-lobe width $2/T_p$. A 3-dB bandwidth and a specified occupied-power bandwidth are different quantities.

A nonzero signal cannot be both strictly time limited and strictly band limited. For a family obtained solely by time scaling a fixed pulse shape, consistently defined duration and bandwidth scale inversely, leaving their product constant. There is no universal fixed duration–bandwidth product for all pulse shapes and definitions.

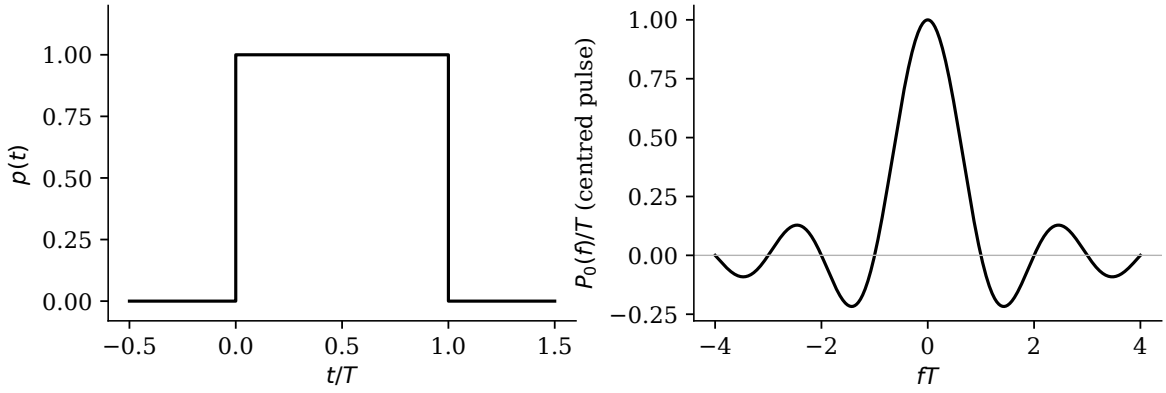


Figure 1.1: A unit-height rectangular pulse of duration T and its spectrum. Time shifting changes spectral phase but not magnitude. The spectral sidelobes extend beyond the first null.

Check your understanding

1. Why does doubling a sinusoid's amplitude multiply its power by four?
2. Find the transform of $3\,\mathrm{rect}((t - t_0)/T)$ without integrating.
3. A pulse is shortened by a factor of two without changing its height. How do energy and first-null bandwidth change?

Answers: power is quadratic; $3T\,\mathrm{sinc}(fT)e^{-j2\pi f t_0}$; energy halves and bandwidth doubles.

Chapter 2

Linear systems, correlation and spectra

Course connection: Week 1 and supplement; Sheets 1–2..

2.1 Impulse response and convolution

Linearity means that scaled sums of inputs produce the same scaled sums of outputs. Time invariance means that delaying the input delays the output by the same amount. These properties imply

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(v)h(t-v)\,\mathrm{d}v, \qquad Y(f) = H(f)X(f). \tag{2.1}$$

Each infinitesimal part of the input excites a shifted impulse response. Convolution adds these contributions. In the frequency domain, an LTI system independently multiplies each frequency by its complex response $H(f)$.

A causal system has $h(t) = 0$ for $t < 0$. For an ordinary impulse-response function, BIBO stability requires $\int |h(t)|\,\mathrm{d}t < \infty$. An ideal rectangular frequency response has an infinitely extended sinc impulse response; it is a mathematical reference rather than a directly implementable causal filter.

Distortionless transmission over the occupied band requires

$$H(f) = Ke^{-j2\pi f t_d}, \tag{2.2}$$

so $y(t) = Kx(t - t_d)$. Flat magnitude alone is insufficient: nonlinear phase causes different spectral components to be delayed differently. Group delay is $\tau_g(f) = -(2\pi)^{-1}\mathrm{d}\arg H/\mathrm{d}f$.

Worked example 2.1 (A triangular frequency response; Sheet 1, Problem 1.30)**.**
Let $h(t) = W\,\mathrm{sinc}^2(Wt)$ and $x(t) = 2 + \cos(\pi Wt) + \cos(2\pi Wt)$.

Solution. The transform pair gives $H(f) = \mathrm{tri}(f/W)$. Evaluate it only at the frequencies in the input: $H(0) = 1$, $H(W/2) = 1/2$ and $H(W) = 0$. Hence

$$y(t) = 2 + \tfrac{1}{2}\cos(\pi Wt).$$

The highest-frequency tone disappears, the middle tone is attenuated, and the DC component passes unchanged.

Worked example 2.2 (Pulse distortion; Sheet 1, Problem 1.32)**.**
Send $x(t) = \mathrm{rect}((t - T/2)/T)$ through $h(t) = ae^{-at}u(t)$.

Solution. For $0 < t < T$, the convolution integrates over $0 < v < t$; for $t > T$, it integrates over $0 < v < T$. Therefore

$$y(t) = \begin{cases} 0, & t < 0, \\ 1 - e^{-at}, & 0 \le t < T, \\ (1 - e^{-aT})e^{-a(t-T)}, & t \ge T. \end{cases} \tag{2.3}$$

The response has finite rise time and a tail after the input ends. Its peak is $1 - e^{-aT}$. Since $H(f) = a/(a + j2\pi f)$, the 3-dB cutoff is $a/(2\pi)$ and the time constant is $1/a$. A short pulse relative to this time constant is heavily rounded.

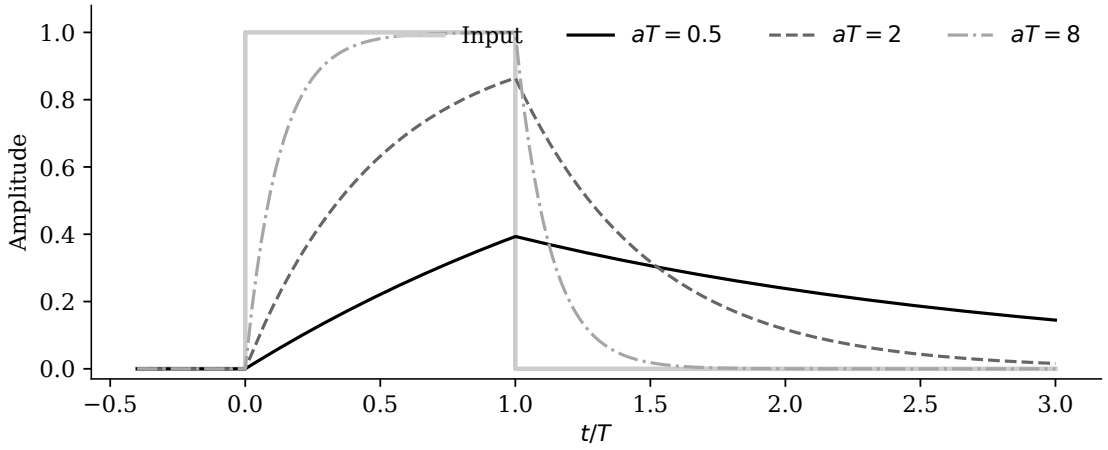


Figure 2.1: The same rectangular pulse through three first-order channels. Larger aT gives a faster response and a higher peak. Tails can interfere with subsequent pulses.

2.2 Correlation measures overlap

For energy signals, fix the convention

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t + \tau)y^*(t)\,\mathrm{d}t. \tag{2.4}$$

Autocorrelation means $y = x$. It satisfies $R_x(0) = E_x$, $R_x(-\tau) = R_x^*(\tau)$, and $|R_x(\tau)| \le R_x(0)$. For real signals it is even. Cross-correlation need not be even or real.

For power signals replace the integral by its long-time average. Then $R_x(0) = P_x$. These are different normalizations and should not be interchanged.

Worked example 2.3 (Correlation of elementary signals; Sheet 2, Problem 2.3)**.**
A unit rectangular pulse has autocorrelation

$$R_p(\tau) = \begin{cases} T - |\tau|, & |\tau| < T, \\ 0, & |\tau| \ge T. \end{cases}$$

This is simply the overlap length of two intervals of length T separated by $|\tau|$.

For $x(t) = e^{-at}u(t)$,

$$R_x(\tau) = \frac{e^{-a|\tau|}}{2a}.$$

For a sinusoid, use the power definition and the product-to-sum identity:

$$R_x(\tau) = \frac{A^2}{2}\cos(2\pi f_0 \tau).$$

Sine and cosine have the same autocorrelation. Autocorrelation does not retain absolute phase.

2.3 Energy density and power density

The correlation theorem gives

$$\mathcal{F}\{R_{xy}\} = X(f)Y^*(f), \qquad \Psi_x(f) = \mathcal{F}\{R_x\} = |X(f)|^2 \tag{2.5}$$

for energy signals. For a power signal, Wiener–Khinchin gives $S_x = \mathcal{F}\{R_x\}$ with the power-normalized correlation. In particular,

$$S_x(f) = \sum_n |c_n|^2 \delta(f - nf_0)$$

for a periodic signal. Integrating an energy density gives energy; integrating a PSD gives power. For normalized electrical signals, PSD units are W/Hz.

A nonzero mean contributes a spectral line $|m_x|^2\delta(f)$. It is the correlation of the fluctuations about the mean, rather than the full correlation, that excludes this DC contribution.

2.4 Filtering correlation and spectral density

For $y = h * x$,

$$S_y(f) = |H(f)|^2 S_x(f), \qquad S_{yx}(f) = H(f)S_x(f), \qquad S_{xy}(f) = H^*(f)S_x(f). \tag{2.6}$$

The same rules hold for energy densities. They are often easier to use than direct time-domain correlation integrals. Compute the output density, then integrate it for power or inverse transform it for correlation.

Worked example 2.4 (Keeping half the energy; Sheet 2, Problem 2.9)**.**
An ideal unit-gain low-pass filter passes $|f| \le B$. For $x(t) = e^{-at}u(t)$,

$$E_y = \int_{-B}^{B} \frac{\mathrm{d}f}{a^2 + (2\pi f)^2} = \frac{1}{\pi a} \arctan \frac{2\pi B}{a}.$$

Since $E_x = 1/(2a)$, imposing $E_y = E_x/2$ gives $B = a/(2\pi)$. The answer follows from integrating the energy spectrum, without calculating the filtered waveform.

Common pitfall. Orthogonality is a zero inner product. Uncorrelated random variables have zero covariance. Independence is a stronger probabilistic property; it does not follow from zero correlation without additional assumptions.

Check your understanding

1. A pure delay has unit magnitude response. What does it do to the PSD and cross-spectrum?
2. Why can two signals have the same autocorrelation but different waveforms?
3. What value should the autocorrelation at zero delay have for a height-A, duration-T pulse?

Answers: PSD unchanged, cross-spectrum gains a phase; spectral phase is not encoded in $|X|^2$; A^2T.

Chapter 3

Sampling, quantization and PCM

Course connection: Week 2; Sheets 3–4 and 12..

3.1 Separating three operations

Sampling makes time discrete. Quantization makes amplitude discrete. Encoding assigns a bit word to each quantized value. In pulse-code modulation (PCM), these operations convert an analog message into a stream of bits. The receiver decodes the words and reconstructs a continuous-time waveform.

Pulse amplitude, width and position modulation (PAM, PWM and PPM) vary a pulse property with the message. They can represent analog values and are not necessarily digital. PCM explicitly represents each quantized sample using a finite bit word. The quantizer is part of analog-to-digital conversion (ADC); a DAC performs the reverse conversion.

3.2 Ideal sampling and spectral replication

With $T_a = 1/f_a$, ideal impulse sampling gives

$$x_\delta(t) = x(t)\sum_n \delta(t - nT_a) = \sum_n x(nT_a)\delta(t - nT_a). \tag{3.1}$$

The impulse train transforms into another impulse train:

$$\sum_n \delta(t - nT_a) \;\leftrightarrow\; \frac{1}{T_a}\sum_k \delta(f - kf_a). \tag{3.2}$$

Multiplication in time becomes convolution in frequency, so

$$X_\delta(f) = \frac{1}{T_a}\sum_k X(f - kf_a). \tag{3.3}$$

Sampling replicates the spectrum at multiples of f_a. It does not simply discard high frequencies.

For $X(f) = 0$ outside $[-B_m, B_m]$, the replicas are separated if $f_a > 2B_m$. The ideal limiting rate is $2B_m$, subject to endpoint conditions. At equality, a sinusoid exactly at the band edge can give ambiguous samples for particular phases. Practical sampling uses a strict margin and an anti-alias low-pass filter before the sampler.

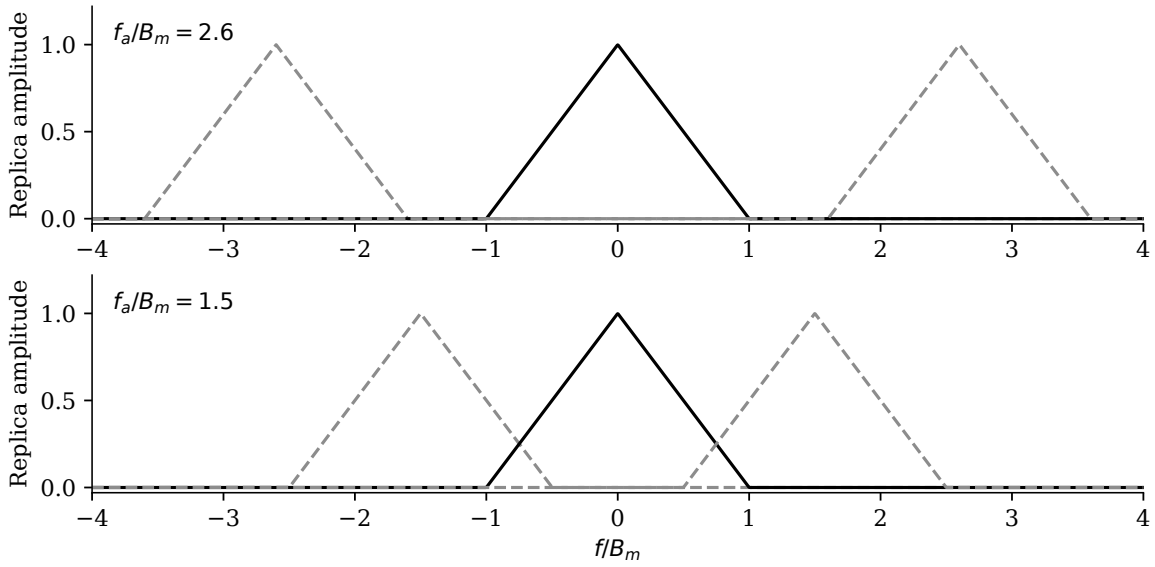


Figure 3.1: Spectral replicas for a triangular baseband spectrum. With $f_a > 2B_m$ they are separated; with $f_a < 2B_m$ they overlap. Vertical scaling is normalized to emphasize the geometry.

Key idea. Aliasing is overlap of spectral replicas. Once distinct analog frequencies produce the same sample sequence, digital processing alone cannot determine which original signal was present.

An ideal reconstruction filter with gain T_a on $|f| < f_a/2$ selects the central replica. Its impulse response is $\mathrm{sinc}(t/T_a)$, giving

$$x(t) = \sum_n x(nT_a)\,\mathrm{sinc}\left(\frac{t - nT_a}{T_a}\right). \tag{3.4}$$

At $t = kT_a$, every term except $n = k$ vanishes. Exact reconstruction requires an infinite sample sequence and an ideal filter; finite implementations approximate it.

Worked example 3.1 (Identifying an alias)**.**
A 7 kHz cosine sampled at 10 ksample/s has samples $\cos(2\pi 0.7n) = \cos(2\pi 0.3n)$. Its samples are identical to those of a 3 kHz cosine. A subsequent low-pass reconstruction produces the alias at 3 kHz.

3.3 Sample-and-hold and linear interpolation

A zero-order hold replaces each sample impulse with a rectangle of duration T_a. Its response is

$$H_{\mathrm{hold}}(f) = T_a\,\mathrm{sinc}(fT_a)e^{-j\pi fT_a}.$$

Together with the $1/T_a$ in Eq. (3.3), this produces baseband amplitude droop $\mathrm{sinc}(fT_a)$ and delay $T_a/2$. Reconstruction must also remove spectral images.

Natural sampling, where a switch gates the actual signal during each aperture, differs from holding the value at a sampling instant. A flat-top sample-and-hold is modeled by convolution of the impulse samples with a rectangle. Confusing these operations gives the wrong spectrum.

Worked example 3.2 (Linear interpolation; Sheet 3, Problem 6.4)**.**

Joining successive samples by straight lines is equivalent to

$$x_0(t) = \sum_n x(nT_a)\,\mathrm{tri}\left(\frac{t - nT_a}{T_a}\right).$$

For $t = nT_a + \lambda T_a$, $0 \le \lambda \le 1$, only two terms contribute: $(1-\lambda)x(nT_a) + \lambda x((n+1)T_a)$. Since the triangle transforms to $T_a\,\mathrm{sinc}^2(fT_a)$,

$$X_0(f) = \mathrm{sinc}^2(fT_a) \sum_k X(f - kf_a).$$

Without aliasing, the central image equals $X(f)\,\mathrm{sinc}^2(fT_a)$. The ideal correction is

$$G(f) = \begin{cases} 1/\,\mathrm{sinc}^2(fT_a), & |f| \le B_m, \\ 0, & \text{otherwise.} \end{cases}$$

The interpolated waveform passes through the samples but is generally not the original waveform. An equalizer corrects its droop only when the sampling process has preserved the information.

3.4 Uniform quantization

A uniform quantizer covering $[-A, A]$ with $L_q = 2^\nu$ intervals has step

$$\Delta = \frac{2A}{L_q}. \tag{3.5}$$

For midpoint reconstruction and no overload, $|e_q| \le \Delta/2$. In the high-resolution model, the error is approximately uniform on $[-\Delta/2, \Delta/2]$, with

$$\mathbb{E}[e_q] \simeq 0, \qquad \sigma_q^2 \simeq \frac{\Delta^2}{12}. \tag{3.6}$$

The variance follows by integrating e^2/Δ over that interval. It is a useful approximation when the input spans many cells and its density changes slowly over a cell. The quantization error is not automatically white or independent of the input; low-level or periodic signals can violate those assumptions. Clipping beyond the quantizer range creates overload distortion, which the uniform-error model omits.

For message power P_m, the quantization SNR is

$$\mathrm{SQNR} \simeq \frac{12P_m}{\Delta^2} = \frac{3P_m}{A^2}2^{2\nu}, \quad \mathrm{SQNR}_{\mathrm{dB}} \simeq 6.0206\nu + 10\log_{10}\frac{3P_m}{A^2}. \tag{3.7}$$

For a full-scale sine, $P_m = A^2/2$, giving $6.02\nu + 1.76$ dB. A symmetric triangular density on $[-A, A]$ has $P_m = A^2/6$, giving $6.02\nu - 3.01$ dB. The offset depends on the amplitude distribution and signal level.

Common pitfall. The familiar $6.02\nu + 1.76$ dB formula is specific to a full-scale sinusoid and an ideal uniform quantizer. It is not a universal ADC SNR.

3.5 Bit budgets and sampling margins

For one uncompressed PCM stream without overhead,

$$R_b = \nu f_a. \tag{3.8}$$

For N identical time-multiplexed channels and b_{oh} framing bits per sampling frame,

$$R_b = f_a(N\nu + b_{\mathrm{oh}}). \tag{3.9}$$

A larger sample rate leaves fewer bits per sample at fixed link rate. Both the sampling constraint and the integer number of bits must be satisfied.

Worked example 3.3 (Voice PCM; Sheet 3, Problem 6.5)**.**
For $B_m = 3.2$ kHz and $R_{b,\max} = 36$ kbit/s, the ideal minimum $f_a = 6.4$ ksample/s permits

$$\nu \le \left\lfloor \frac{36000}{6400} \right\rfloor = 5, \qquad L_q = 32.$$

Five bits at this sample rate use 32 kbit/s. The same word length can run up to $f_a = 7.2$ ksample/s, allowing a sampling margin. Six bits would require at least 38.4 kbit/s.

Worked example 3.4 (Resolution and maximum error; Sheet 3, Problem 6.6)**.**
For $B_m = 10$ kHz, $A = 5$ V and binary signaling limited to 150 kbaud, $R_{b,\max} = 150$ kbit/s. At the ideal $f_a = 20$ ksample/s,

$$\nu = 7, \quad L_q = 128, \quad R_b = 140 \text{ kbit/s}.$$

At full link utilization, $f_a = 150000/7 = 21.429$ ksample/s. The step is $10/128 = 0.078125$ V, so the maximum granular error is 0.0390625 V.

Worked example 3.5 (A triangular input and a guard band; Sheet 3, Problem 6.11)**.**
Let $p_X(x) = \frac{1}{2}\,\mathrm{tri}(x/2)$ and $B_m = 5$ kHz. Direct integration gives $P_m = 2/3$, while $\Delta = 4/L_q$. Therefore SQNR $\simeq L_q^2/2$.

With $L_q = 32$ and $f_a = 10$ ksample/s, $R_b = 50$ kbit/s and SQNR is 27.09 dB. A 40 kbit/s rate limit reduces the maximum word length to four bits, giving 21.07 dB.

If a 2 kHz gap is required between adjacent spectral replicas, then $f_a - 2B_m \ge 2$ kHz, or $f_a \ge 12$ ksample/s. The rate limit now permits only three bits per sample, giving 15.05 dB. The guard band costs one bit of resolution in this example.

3.6 Companding

Many useful inputs occur more often at small amplitudes. A compressor expands the small-amplitude region before uniform quantization; the receiver applies an inverse expander. The combination implements nonuniform quantization in the original amplitude variable.

For normalized input $u = m/A \in [-1, 1]$, the μ-law characteristic is

$$v = \mathrm{sgn}(u)\frac{\ln(1+\mu|u|)}{\ln(1+\mu)}, \qquad u = \mathrm{sgn}(v)\frac{(1+\mu)^{|v|}-1}{\mu}. \tag{3.10}$$

The left-hand variable in the inverse formula is u (normalized amplitude), not the bit count ν. Small input amplitudes receive finer effective steps because the compressor slope is largest near zero.

For the A-law parameter $A_c > 1$, use a separate symbol to avoid confusing it with peak voltage:

$$v = \mathrm{sgn}(u)\begin{cases} \dfrac{A_c|u|}{1+\ln A_c}, & 0 \le |u| < 1/A_c, \\ \dfrac{1+\ln(A_c|u|)}{1+\ln A_c}, & 1/A_c \le |u| \le 1. \end{cases} \tag{3.11}$$

The mapping is continuous at the breakpoint. Companding redistributes quantization error; it does not increase the number of bits.

Worked example 3.6 (Exact effective steps; Sheet 4, Problem 6.12)**.**
With $A = 40$ V, $L_q = 256$ and $\mu = 255$, the uniform step without compression is 0.3125 V. Define

$$g(z) = \operatorname{sgn}(z)\frac{40}{255}\left(256^{|z|/40} - 1\right).$$

If quantizer reconstruction values in the compressed variable are cell midpoints separated by $\Delta_z = 80/256$, then the smallest and largest spacings between reconstructed input levels are

$$d_{\min} = g(\Delta_z/2) - g(-\Delta_z/2) \simeq 6.87 \text{ mV},$$
$$d_{\max} = g(40 - \Delta_z/2) - g(40 - 3\Delta_z/2) \simeq 1.666 \text{ V}.$$

A cell width between decision boundaries is a different quantity. For example, the outermost width is $g(40) - g(40 - \Delta_z) \simeq 1.702$ V. State which spacing is being calculated.

Check your understanding

1. Does doubling the sampling rate improve quantization resolution at fixed R_b?
2. How many uniform quantization levels are needed for $|e_q| \leq 0.01A$?
3. Can a reconstruction filter undo aliasing?

Answers: it reduces the available bits per sample; at least 100 levels, hence 128 for binary PCM; not without additional information about the input.

Chapter 4

Line coding, pulse shaping and ISI

Course connection: Week 2; Sheets 4, 6 and 12..

4.1 The PAM model

A common digital waveform model is

$$x(t) = \sum_{n=-\infty}^{\infty} a_n p(t - nT_s). \tag{4.1}$$

The sequence selects the symbol amplitudes, and $p(t)$ determines their waveform. In binary signaling, $M = 2$ and $R_b = R_s$. For fixed-length $M = 2^k$ signaling, each symbol represents k bits, so $R_b = kR_s$.

Non-return-to-zero (NRZ) pulses occupy the full symbol interval. Return-to-zero (RZ) pulses occupy only a fraction of it. Polar binary coding uses $\pm A$; unipolar coding uses $0, A$. Manchester coding uses opposite signs in the two halves of every bit interval. Alternate mark inversion (AMI) represents zeros by zero amplitude and alternates the signs of successive ones.

Code	Useful property	Limitation
Unipolar NRZ	Simple two-level waveform	DC component, poor timing on long runs
Polar NRZ	Zero mean for balanced data	Long runs still lack transitions
Polar RZ	More transitions for timing	Wider spectrum than NRZ
Manchester	A transition within every bit; zero pulse area	Larger first-null bandwidth
AMI	Suppressed DC; violations can flag errors	Long runs of zeros still need treatment

These are waveform properties. They do not by themselves provide a general error-correcting code.

4.2 Power spectral density of random PAM

Fixed-clock random PAM is generally cyclostationary. The following expressions describe its time-averaged PSD, or the WSS model obtained by uniformly randomizing the timing origin over one symbol interval.

Let $R_a[\ell] = \mathbb{E}[a_{n+\ell}a_n^*]$. Then

$$S_x(f) = \frac{|P(f)|^2}{T_s} \sum_{\ell=-\infty}^{\infty} R_a[\ell] e^{-j2\pi f \ell T_s}. \tag{4.2}$$

For independent identically distributed amplitudes with mean μ_a and variance σ_a^2, $R_a[\ell] = \sigma_a^2 \delta_{\ell 0} + |\mu_a|^2$. Using the Fourier comb identity,

$$S_x(f) = \frac{\sigma_a^2}{T_s}|P(f)|^2 + \frac{|\mu_a|^2}{T_s^2} \sum_k |P(k/T_s)|^2 \delta(f - k/T_s). \tag{4.3}$$

Random fluctuations produce the continuous spectrum; a periodic mean waveform produces spectral lines. Independence alone does not make $R_a[\ell]$ zero for $\ell \neq 0$: zero mean is also required.

For zero-mean independent amplitudes,

$$P_x = \frac{\mathbb{E}[|a|^2]E_p}{T_s}, \qquad E_p = \int |p(t)|^2 \, \mathrm{d}t. \tag{4.4}$$

The expression also holds for nonoverlapping pulses with nonzero mean. For general overlapping pulses and correlated or nonzero-mean symbols, cross terms may contribute, and Eq. (4.2) is safer.

Worked example 4.1 (Comparing line-code spectra; Sheet 4, Problem 6.13)**.**
For a unit NRZ pulse, $|P(f)|^2 = T_s^2 \operatorname{sinc}^2(fT_s)$. Equiprobable polar amplitudes $\pm A$ give

$$S_{\text{polar}}(f) = A^2 T_s \operatorname{sinc}^2(fT_s).$$

Equiprobable unipolar amplitudes $0, A$ have mean $A/2$ and variance $A^2/4$, giving

$$S_{\text{uni}}(f) = \frac{A^2 T_s}{4} \operatorname{sinc}^2(fT_s) + \frac{A^2}{4}\delta(f).$$

Only the DC line survives because the NRZ spectrum vanishes at nonzero multiples of $1/T_s$. The total powers are A^2 and $A^2/2$, respectively. Thus equal amplitudes do not mean equal bit energies.

For Manchester, take $p = +1$ on the first half and $p = -1$ on the second. Then

$$|P_M(f)|^2 = T_s^2 \operatorname{sinc}^2(fT_s/2) \sin^2(\pi f T_s/2).$$

The DC null follows from the zero pulse area. Its energy remains T_s, so the polar Manchester and polar NRZ codes have the same energy for the same amplitude and duration.

For a fair unit-average-bit-energy comparison, choose $A = 1/\sqrt{T_s}$ for polar NRZ and Manchester, $A = \sqrt{2/T_s}$ for unipolar NRZ or polar half-duty RZ, and $A = 2/\sqrt{T_s}$ for unipolar half-duty RZ.

4.3 Intersymbol interference

Let $g(t)$ be the overall pulse at the sampler, including transmitter filtering, channel and receiver filtering. The sample at kT_s is

$$z_k = a_k g(0) + \underbrace{\sum_{n \neq k} a_n g((k-n)T_s)}_{\text{ISI}} + w_k. \tag{4.5}$$

ISI depends on surrounding data. Noise remains random even if the data are known. A filter may reduce noise yet increase ISI if it changes the overall pulse at the decision instants.

With normalization $g(0) = 1$, zero ISI requires

$$g(kT_s) = \delta_{k0}. \tag{4.6}$$

Pulses may overlap strongly between sample times and still satisfy this condition. The frequency-domain Nyquist criterion is

$$\sum_{n=-\infty}^{\infty} G(f - n/T_s) = T_s. \tag{4.7}$$

A minimum-bandwidth solution is $g(t) = \operatorname{sinc}(t/T_s)$ with $B_{\mathrm{BB}} = 1/(2T_s)$. Hence ideal zero-ISI real baseband signaling supports $R_s \le 2B_{\mathrm{BB}}$.

4.4 Raised-cosine pulses

The sinc has slowly decaying tails and is sensitive to finite truncation and timing errors. A raised-cosine spectrum allows excess bandwidth to obtain faster decay:

$$G(f) = \begin{cases} T_s, & |f| \le (1-\alpha)/(2T_s), \\ \dfrac{T_s}{2}\left[1 + \cos\left(\dfrac{\pi T_s}{\alpha}\left(|f| - \dfrac{1-\alpha}{2T_s}\right)\right)\right], & \dfrac{1-\alpha}{2T_s} < |f| \le \dfrac{1+\alpha}{2T_s}, \\ 0, & \text{otherwise.} \end{cases} \tag{4.8}$$

For $0 < \alpha \le 1$, the impulse response is

$$g(t) = \operatorname{sinc}(t/T_s)\frac{\cos(\pi\alpha t/T_s)}{1 - (2\alpha t/T_s)^2}, \tag{4.9}$$

with the removable singularities defined by continuity. At the singular points,

$$g(\pm T_s/(2\alpha)) = (\pi/4)\operatorname{sinc}(1/(2\alpha)).$$

The bandwidth is

$$B_{\mathrm{BB}} = \frac{1+\alpha}{2}R_s. \tag{4.10}$$

Usually transmitter and matched receiver share this response: each uses a root-raised-cosine filter, so their product is raised cosine for a flat channel. Zero ISI is a property of the overall response. A single root-raised-cosine pulse need not vanish at all other symbol instants.

Worked example 4.2 (Bandwidth and rate; Sheet 4, Problems 6.20–6.21)**.**
At $R_b = R_s = 100$ kbit/s, roll-off values 0, 0.5 and 1 require baseband bandwidths 50, 75 and 100 kHz. Conversely, $B_{\mathrm{BB}} = 3.5$ kHz and $\alpha = 0.25$ permit

$$R_s = \frac{2(3500)}{1.25} = 5600 \text{ baud}.$$

The bit rate is 5600 bit/s only for binary signaling; for M-ary signaling it is $5600 \log_2 M$ bit/s, before coding overhead.

Worked example 4.3 (Telemetry design; Sheet 6, Problem 6.23)**.**
Eight messages each have $B_m = 2$ kHz. A 25% sampling margin requires $f_a = 1.25(2B_m) = 5$ ksample/s per message. A maximum quantization error of 1% of peak requires $A/L_q \le 0.01A$, so $L_q \ge 100$ and $\nu = 7$.

$$R_b = 8(5000)(7) = 280 \text{ kbit/s}, \qquad B_{\mathrm{BB}} = \frac{1.2}{2}(280 \text{ kbaud}) = 168 \text{ kHz}.$$

This result assumes no framing or channel-coding overhead.

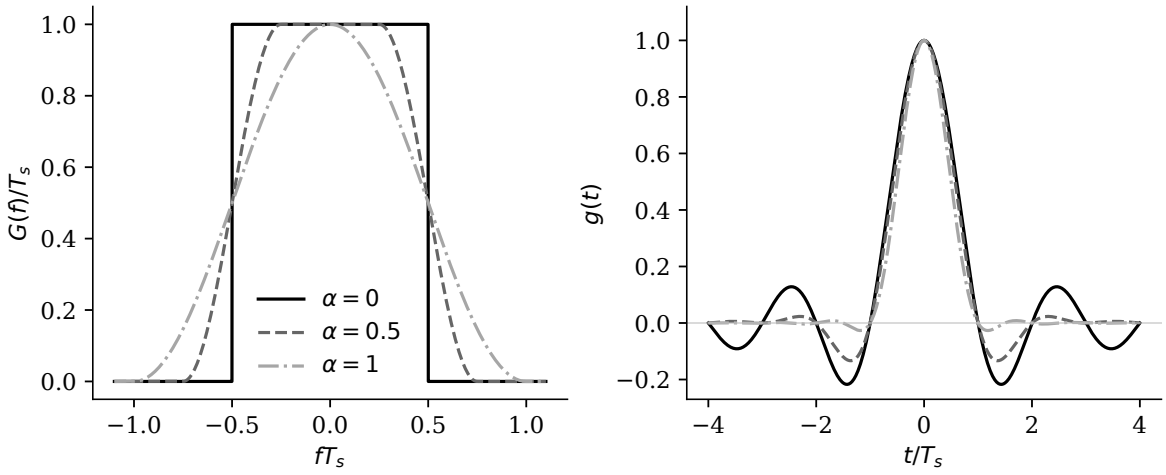


Figure 4.1: Raised-cosine spectra and overall pulses for three roll-off values. All pulses equal one at zero and vanish at the other integer symbol times. More excess bandwidth reduces the long tails.

Worked example 4.4 (Resolution versus roll-off; Sheet 12, Problem 6.24)**.**
A binary baseband link has $B_{\mathrm{BB}} = 192$ kHz and $\alpha = 0.5$, so $R_b \le 256$ kbit/s. The triangular input of Chapter 3 requires $\nu = 8$ to reach 45 dB SQNR. Thus the ideal maximum message bandwidth is

$$B_m = \frac{R_b}{2\nu} = 16 \text{ kHz}.$$

Keeping this message bandwidth but reducing α to 0.2 raises the available bit rate to 320 kbit/s. At $f_a = 32$ ksample/s, ten bits per sample fit, increasing SQNR from 45.15 to 57.20 dB. The improvement requires actually using the extra rate for extra quantizer bits.

Worked example 4.5 (User count and bandwidth convention; Sheet 6, Problem 6.25)**.**
The message PSD is $2\times10^{-5}\,\mathrm{tri}(f/(2\times10^4))$ W/Hz. Its area gives $P_m = 0.4$ W and its bandwidth is 20 kHz. With peak amplitude one, SQNR $\simeq 1.2L_q^2$, so 45 dB requires eight bits. Each user generates $40{,}000 \times 8 = 320$ kbit/s at ideal Nyquist sampling.

Using the sheet's NRZ *first-null* convention for a 3 MHz baseband channel gives approximately 3 Mbaud. Binary signaling supports $\lfloor 3/0.320 \rfloor = 9$ users, and octal signaling supports $\lfloor 9/0.320 \rfloor = 28$ users. This is a main-lobe bandwidth estimate, not a strict claim that rectangular pulses pass without distortion. A zero-ISI raised-cosine design must instead use Eq. (4.10).

4.5 General equally spaced PAM levels

For $a_m = [m - (M+1)/2]d$, $m = 1, \ldots, M$, equiprobability gives

$$\mathbb{E}[a] = 0, \qquad \mathbb{E}[a^2] = \frac{d^2(M^2-1)}{12}.$$

With unit-height NRZ pulses, this is the average power. At fixed nearest-neighbor spacing d, increasing M costs power; at fixed power, spacing decreases and noise errors become more likely.

In Sheet 12, Problem 6.14, the lower half of the levels have probabilities $2/(3M)$ and the upper half have probabilities $4/(3M)$. Pairing opposite levels shows that the second moment is unchanged, while

$$\mu_a = \frac{Md}{12}, \qquad \sigma_a^2 = \frac{d^2(11M^2-12)}{144}.$$

For NRZ, the continuous spectrum integrates to σ_a^2 and the DC line has weight μ_a^2. Their sum is the original power.

Check your understanding

1. Why does zero mean eliminate spectral lines for independent PAM amplitudes?
2. A 16-level link operates at 2 Mbit/s with $\alpha = 0.25$. Find R_s and B_{BB}.
3. Is a wide-open eye between sample times required by the zero-ISI condition?

Answers: the periodic mean term vanishes; 0.5 Mbaud and 0.3125 MHz; the exact criterion concerns the sample instants, although an opening around them provides timing tolerance.

Chapter 5

Amplitude modulation and coherent reception

Course connection: Week 3; Sheet 5..

5.1 Why translate a spectrum?

Modulation places a message in a frequency band suited to the physical channel. It also allows different users to occupy different carrier bands. Multiplication by a cosine shifts a baseband spectrum to positive and negative carrier frequencies according to Eq. (1.11). The negative-frequency copy is required by the reality of the waveform; it is not a second independently allocated radio channel.

Throughout this chapter, $m(t)$ is real, zero mean, bandlimited to B_m, and has power P_m. We assume the carrier frequency is high enough to separate the translated spectra.

5.2 Conventional AM

Write the transmitted signal as

$$s(t) = A_c[m_{\mathrm{dc}} + m(t)]\cos(2\pi f_c t). \tag{5.1}$$

The offset m_{dc} produces the transmitted carrier. The spectrum is

$$\begin{aligned} S_{\mathrm{amp}}(f) = {} & \frac{A_c m_{\mathrm{dc}}}{2}[\delta(f - f_c) + \delta(f + f_c)] \\ & + \frac{A_c}{2}[M(f - f_c) + M(f + f_c)]. \end{aligned} \tag{5.2}$$

Here S_{amp} is a waveform transform, not a PSD. The positive-frequency occupied band runs from $f_c - B_m$ to $f_c + B_m$, so $B_T = 2B_m$.

The RF envelope is $A_c|m_{\mathrm{dc}} + m(t)|$. Envelope detection can reproduce the offset message without folding only if $m_{\mathrm{dc}} + m(t) \geq 0$. For a symmetric peak bound $|m(t)| \leq m_p$, $m_{\mathrm{dc}} \geq m_p$ is sufficient. Define $\mu = m_p/m_{\mathrm{dc}}$; the usual no-overmodulation condition is $\mu \leq 1$.

5.3 Power and efficiency

Averaging Eq. (5.1) squared gives

$$P_T = \frac{A_c^2}{2}(m_{\mathrm{dc}}^2 + P_m), \qquad P_c = \frac{A_c^2 m_{\mathrm{dc}}^2}{2}, \qquad P_{\mathrm{SB}} = \frac{A_c^2 P_m}{2}. \tag{5.3}$$

The cross term vanishes because the message has zero mean. The sideband efficiency is

$$\eta_{\mathrm{AM}} = \frac{P_{\mathrm{SB}}}{P_T} = \frac{P_m}{m_{\mathrm{dc}}^2 + P_m}. \tag{5.4}$$

For a sinusoidal message with peak m_p, $P_m = m_p^2/2$, and $\eta_{\mathrm{AM}} = \mu^2/(2+\mu^2)$. For this single-tone case, $\mu \leq 1$ limits efficiency to 1/3. The one-third bound is not universal for every message waveform.

Worked example 5.1 (Single-tone AM; Sheet 5, Problem 3.6)**.**
For $s(t) = 10[1 + 2k\cos(2\pi f_0 t)]\cos(2\pi f_c t)$ and $k = 0.25$, the modulation depth is $\mu = 0.5$. Thus $P_c = 50$, $P_{\mathrm{SB}} = 6.25$, $P_T = 56.25$ in normalized watts, and $\eta = 1/9 = 11.11\%$.

For $k = 1.25$, the factor $1 + 2.5\cos(2\pi f_0 t)$ changes sign. The RF waveform remains a valid AM waveform, but an envelope detector folds the negative portions and cannot recover the original sinusoid linearly.

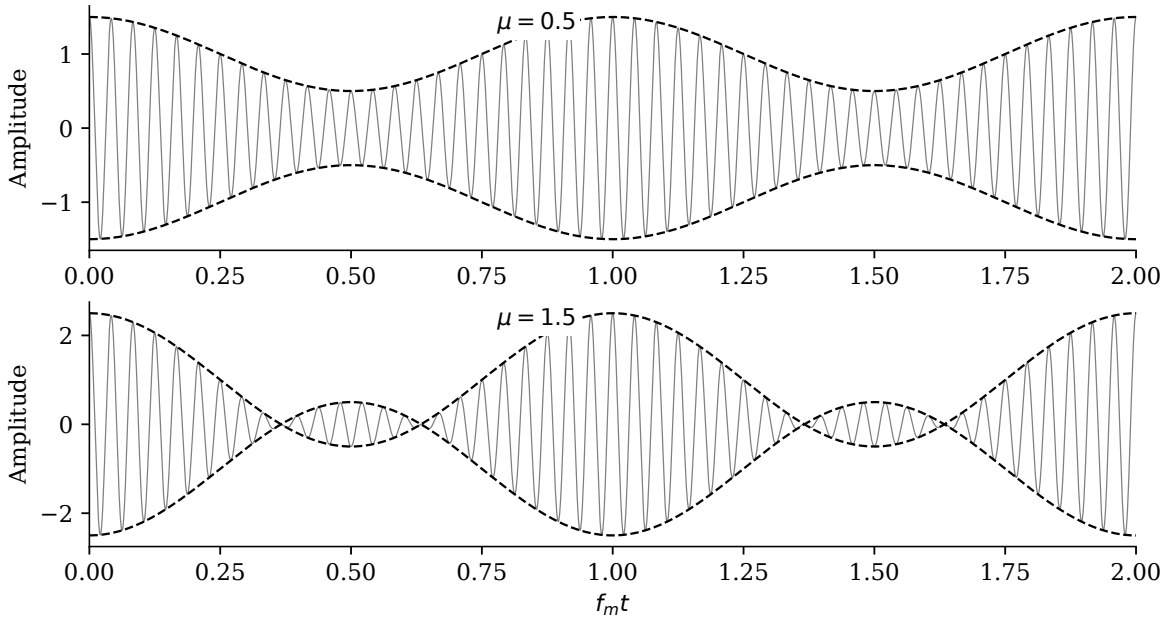


Figure 5.1: AM with modulation depths 0.5 and 1.5. The dashed curves show the signed amplitude and its negative. In overmodulation, the positive envelope folds at the zero crossings of the signed amplitude.

5.4 Envelope detection and rectification

An ideal envelope detector returns the RF envelope. A diode-capacitor detector approximates it when its discharge time is long compared with a carrier period and short compared with relevant message variation times. A rough guide is $1/f_c \ll RC \ll 1/B_m$, but avoiding diagonal clipping also depends on the steepest envelope slope.

Full-wave rectification followed by low-pass filtering is a related but differently scaled operation. Since

$$\langle |\cos(2\pi f_c t)| \rangle = \frac{2}{\pi},$$

non-overmodulated AM produces

$$\mathrm{LPF}\{|s(t)|\} = \frac{2A_c}{\pi}[m_{\mathrm{dc}} + m(t)] \tag{5.5}$$

when the low-pass filter passes the message and rejects the rectification images. Remove DC and multiply by $\pi/(2A_c)$ to recover $m(t)$.

One can also derive this using $q(t) = \operatorname{sgn}\cos(2\pi f_c t)$ and $|s| = sq$. The Fourier coefficients satisfy $c_0 = 0$ and $c_k = \operatorname{sinc}(k/2)$ for $k \neq 0$. Both $k = +1$ and $k = -1$ produce a baseband copy. Keeping only one loses a factor of two.

Worked example 5.2 (Rectifier gain; Sheet 5, Problems 3.4 and 3.6)**.**
For the preceding $k = 0.25$ signal, rectification and low-pass filtering give $20[1 + 0.5\cos(2\pi f_0 t)]/\pi$. After DC removal the cosine amplitude is $10/\pi$. To obtain $2\cos(2\pi f_0 t)$, choose gain $G = \pi/5$.

With $f_0 = 1$ kHz and $f_c = 10$ kHz, a 2 kHz low-pass filter passes the message and rejects the nearest rectification image around $2f_c$. Under overmodulation, the output must be calculated by filtering $|1 + 2k\cos(2\pi f_0 t)|$; a finite-bandwidth filter need not retain all harmonics of that folded envelope.

5.5 Suppressed carrier and coherent detection

Double-sideband suppressed-carrier (DSB-SC) modulation sets $m_{\mathrm{dc}} = 0$:

$$s(t) = A_c m(t)\cos(2\pi f_c t), \qquad P_T = A_c^2 P_m/2.$$

It retains the $2B_m$ bandwidth but removes the carrier-power cost. Because the signed message can be negative, envelope detection loses its sign.

A coherent receiver multiplies by $2\cos(2\pi f_c t + \theta)$. Product-to-sum expansion followed by low-pass filtering gives

$$y(t) = A_c[m_{\mathrm{dc}} + m(t)]\cos\theta. \tag{5.6}$$

For constant $\theta = 0$, the message is recovered exactly after gain normalization and DC removal, even if the original AM was overmodulated. At $\theta = \pi/2$ the desired component disappears. A frequency error makes $\theta(t) = 2\pi\Delta f_{\mathrm{LO}} t + \theta_0$, causing time-dependent fading or sign reversals.

Common pitfall. If θ varies in time, the term $A_c m_{\mathrm{dc}}\cos\theta(t)$ is not DC. An AC-coupling capacitor does not automatically remove the entire term. Its rejection depends on the actual high-pass response. This matters in Sheet 5, Problem 3.10.

5.6 Single-sideband modulation

For a real message, the two AM sidebands carry redundant information. Let $\widehat{m}(t)$ be its Hilbert transform, with frequency response $-j\operatorname{sgn} f$. An upper-sideband signal is

$$s_{\mathrm{USB}}(t) = A_c[m(t)\cos(2\pi f_c t) - \widehat{m}(t)\sin(2\pi f_c t)]. \tag{5.7}$$

Changing the minus sign to plus gives the lower sideband. For a single-tone message, $\widehat{m}(t) = \sin(2\pi f_m t)$ and the USB expression becomes $A_c\cos[2\pi(f_c + f_m)t]$, confirming the sign convention.

SSB requires only $B_T = B_m$. With the amplitude normalization above, its power is $A_c^2 P_m$, since the two quadrature terms each contribute $A_c^2 P_m/2$. This differs from the DSB-SC normalization; compare systems at equal transmitted power, not merely equal A_c. Coherent detection recovers $A_c m(t)$ after multiplication by $2\cos(2\pi f_c t)$ and low-pass filtering.

5.7 Quadrature modulation and frequency conversion

The I/Q representation used here is

$$s(t) = I(t)\cos(2\pi f_c t) - Q(t)\sin(2\pi f_c t) = \Re\{[I(t) + jQ(t)]e^{j2\pi f_c t}\}. \tag{5.8}$$

Coherent multiplication by $2\cos$ and $-2\sin$, followed by low-pass filtering, recovers I and Q. They may carry independent messages because the carriers are orthogonal under the assumed filtering. In analog QAM each message of bandwidth B_m shares a total passband width $2B_m$.

A superheterodyne receiver mixes the chosen RF channel to a fixed intermediate frequency. If $f_{\mathrm{LO}} = f_c + f_{\mathrm{IF}}$, then an unwanted input at $f_c + 2f_{\mathrm{IF}}$ also converts to the same IF. This is the image frequency. The RF filter must reject it before mixing, while the fixed IF filter provides channel selectivity. Mixing translates every admitted signal, including interference.

Check your understanding

1. What fraction of the power carries the message in single-tone AM at $\mu = 1$?
2. Why does a 180° coherent-receiver phase error invert the recovered message?
3. How does SSB reduce bandwidth without discarding information in a real message?

Answers: one third; $\cos\pi = -1$; the two DSB sidebands are redundant.

Chapter 6

Frequency and phase modulation

Course connection: Week 4; Sheet 7..

6.1 Instantaneous phase and frequency

Write an angle-modulated waveform as

$$s(t) = A_c \cos\theta(t), \qquad \theta(t) = 2\pi f_c t + \phi(t). \tag{6.1}$$

Its instantaneous frequency is

$$f_i(t) = \frac{1}{2\pi}\frac{\mathrm{d}\theta}{\mathrm{d}t} = f_c + \frac{1}{2\pi}\frac{\mathrm{d}\phi}{\mathrm{d}t}. \tag{6.2}$$

The envelope is constant. In the usual carrier-averaging regime, the transmitted power is $A_c^2/2$ and does not increase with modulation depth. This facilitates efficient RF amplification but does not mean that every spectral component separately contains recoverable information.

We use the following sensitivity conventions:

$$\text{PM:}\quad \phi(t) = k_p m(t), \qquad f_i(t) = f_c + \frac{k_p}{2\pi} m'(t), \tag{6.3}$$

$$\text{FM:}\quad \phi(t) = 2\pi k_f \int_{t_0}^{t} m(v)\,\mathrm{d}v + \phi_0, \qquad f_i(t) = f_c + k_f m(t). \tag{6.4}$$

Thus k_p is in rad per message unit and k_f in Hz per message unit. If a source defines FM sensitivity in rad/s per unit, divide that sensitivity by 2π before using these formulas.

An FM modulator can implement PM by differentiating the message with the appropriate gain. A PM modulator can implement FM by integrating it. The phase integration constant affects carrier phase, not instantaneous frequency.

Worked example 6.1 (Reading an FM waveform; Sheet 7, Problem 3.19)**.**
Use $\cos a \cos b + \sin a \sin b = \cos(a - b)$ to write the given signal as

$$s(t) = \cos[200\pi t - 10\sin(2\pi t)].$$

With $f_c = 100$ Hz, the phase deviation is $-10\sin(2\pi t)$ rad and

$$f_i(t) = 100 - 10\cos(2\pi t) \text{ Hz}.$$

A normalized message is $m(t) = -\cos(2\pi t)$ with $k_f = 10$ Hz per unit amplitude. Under Eq. (5.8), $I = \cos[10\sin(2\pi t)]$ and $Q = -\sin[10\sin(2\pi t)]$. The sheet uses a plus-sine representation and therefore the opposite sign for Q.

A different reference carrier can formally be absorbed into a linear term in $\phi(t)$. The zero-mean-message convention selects the central carrier naturally.

6.2 A sinusoidal message and modulation index

For $m(t) = A_m \cos(2\pi f_m t)$, FM gives

$$s(t) = A_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)], \quad \Delta f = k_f A_m, \quad \beta = \frac{\Delta f}{f_m}. \tag{6.5}$$

For PM the peak phase deviation is $k_p A_m$ and the peak frequency deviation is $k_p A_m f_m$. Therefore doubling message frequency at fixed amplitude leaves FM peak frequency deviation unchanged, but doubles the PM peak frequency deviation.

A discontinuous message produces discontinuous PM phase, whose derivative includes impulses. It has no finite global peak frequency deviation in this idealization. In FM, a jump in a bounded message changes the frequency abruptly but the accumulated phase stays continuous.

Worked example 6.2 (A piecewise message; Sheet 7, Problem 3.21)**.**
For the sheet's message, $\max|m| = 1$ and the maximum absolute accumulated area is $3/2$ in amplitude–time units. The maximum PM phase deviation is k_p, while the FM value is $2\pi k_f(3/2)$. Equal peak phase deviations require $k_p = 3\pi k_f$, with the time and amplitude units understood.

For $k_f = 1$ Hz/V and peak amplitude 1 V, FM has peak deviation 1 Hz. For $k_p = 1$ rad/V, the unit-slope smooth segment gives $1/(2\pi)$ Hz PM deviation. However, the jumps in the message generate frequency impulses; $1/(2\pi)$ Hz is only the maximum over the smooth segments, not over the entire ideal waveform.

6.3 Narrowband and wideband spectra

For small phase deviation $|\phi(t)| \ll 1$,

$$s(t) \simeq A_c \cos(2\pi f_c t) - A_c \phi(t) \sin(2\pi f_c t).$$

This narrowband approximation has a carrier plus two sidebands, approximately occupying $2B_m$. It is the small *phase deviation*, rather than merely $B_m \ll f_c$, that justifies linearization.

For a single FM tone, the exact expansion uses Bessel functions:

$$e^{j\beta \sin(2\pi f_m t)} = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi n f_m t}. \tag{6.6}$$

There are spectral lines at $f_c + n f_m$ with signed amplitudes $A_c J_n(\beta)$ in the cosine expansion. The sidebands extend indefinitely, although sufficiently distant terms are small. The identity $\sum_n J_n^2(\beta) = 1$ expresses the redistribution of constant total power among the sidebands.

Carson's practical occupied-bandwidth estimate is

$$B_T \simeq 2(\Delta f + B_m). \tag{6.7}$$

For a single tone this becomes $2(\beta + 1) f_m$. It is an engineering estimate, not a strict spectral cutoff. In the wideband regime $\Delta f \gg B_m$, the leading estimate is $B_T \simeq 2\Delta f$.

6.4 Quasi-stationary spectral approximation

When the message varies slowly compared with the carrier and the deviation is large compared with message bandwidth, one can approximate a smoothed FM spectrum using the distribution of instantaneous frequency. If the message amplitude has density p_m and $k_f > 0$,

$$S_s(f) \simeq \frac{A_c^2}{4k_f}\left[p_m\left(\frac{f-f_c}{k_f}\right) + p_m\left(\frac{-f-f_c}{k_f}\right)\right]. \tag{6.8}$$

Each translated density integrates to $A_c^2/4$. For a tone observed uniformly over its period,

$$p_m(u) = \frac{1}{\pi\sqrt{A_m^2 - u^2}}, \qquad |u| < A_m. \tag{6.9}$$

This arcsine density follows by transforming a uniform phase variable and summing the two inverse branches of the cosine. For deterministic periodic FM the exact spectrum remains a set of Bessel lines. Equation (6.8) describes their coarse spectral envelope, not an exact continuous replacement.

Worked example 6.3 (Wideband FM; Sheet 7, Problem 3.24)**.**
For $f_c = 1$ MHz, $m(t) = 2\cos(200\pi t)$ and $k_f = 3000$ Hz/V, $f_m = 100$ Hz and $\Delta f = 6000$ Hz. Here $f_c \gg \Delta f \gg f_m$ and $\beta = 60$, consistent with the wideband quasi-stationary approximation.

The leading bandwidth estimate is 12 kHz; Carson gives 12.2 kHz. Halving message amplitude makes $\beta = 30$, gives a 6 kHz leading estimate, and a 6.2 kHz Carson estimate. Checking only $f_m \ll f_c$ would not establish the large-deviation condition.

6.5 Frequency demodulation

Differentiating the RF waveform gives

$$\frac{\mathrm{d}s}{\mathrm{d}t} = -2\pi A_c f_i(t)\sin\theta(t).$$

If the instantaneous frequency remains positive, its envelope is $2\pi A_c f_i(t)$. Removing the carrier-frequency term gives a signal proportional to $m(t)$. An amplitude limiter before a practical discriminator suppresses amplitude fluctuations, although it cannot eliminate phase noise.

An I/Q receiver obtains $I = A_c\cos\phi$ and $Q = A_c\sin\phi$. It estimates the phase using $\mathrm{atan2}(Q, I)$, unwraps it, then differentiates:

$$\widehat{m}(t) = \frac{1}{2\pi k_f}\frac{\mathrm{d}}{\mathrm{d}t}\,\mathrm{unwrap}\{\mathrm{atan2}(Q, I)\}. \tag{6.10}$$

A fixed local-oscillator phase offset disappears upon differentiation. A frequency offset does not: it adds a constant to the demodulated message.

Common pitfall. The I/Q low-pass filters must pass the complex-envelope spectrum, approximately $\Delta f + B_m$ by Carson's estimate, not merely B_m. The nonlinear functions $\cos\phi(t)$ and $\sin\phi(t)$ can be much wider than the message. Also, $\arctan(Q/I)$ without quadrant handling and unwrapping is not the full phase.

Check your understanding

1. A tone has $f_m = 2$ kHz and $\Delta f = 10$ kHz. Find β and Carson bandwidth.
2. What happens to FM power when the deviation doubles at fixed carrier amplitude?
3. Why is differentiating noisy phase measurements difficult?

Answers: $\beta = 5$ and $B_T \simeq 24$ kHz; power stays approximately constant; differentiation amplifies high-frequency noise.

Chapter 7

Random processes and noise

Course connection: Week 5; Sheet 8..

7.1 Ensembles and realizations

A random process $X(t)$ associates a random variable with every time. One experiment gives a realization $x(t)$; the ensemble describes all possible realizations and their probabilities. The mean and autocorrelation are

$$\mu_X(t) = \mathbb{E}[X(t)], \qquad R_X(t_1, t_2) = \mathbb{E}[X(t_1)X^*(t_2)]. \tag{7.1}$$

The covariance subtracts the means: $C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X^*(t_2)$. At equal times it gives the variance.

A process is strictly stationary if every finite-dimensional joint distribution is invariant under a common time shift. It is wide-sense stationary (WSS) if its second moments are finite, its mean is constant and its autocorrelation depends only on $\tau = t_1 - t_2$.

Ergodicity is a separate property: an appropriate time average from a realization reproduces the ensemble quantity. WSS alone does not imply it. For example, $X(t) = Z$, where Z is a random constant with nonzero variance, is stationary. The time average of any realization is its particular value Z, not necessarily $\mathbb{E}[Z]$.

7.2 A random-phase sinusoid

Worked example 7.1 (Uniform phase; Sheet 8, Problem 4.18)**.**
Let $X(t) = A_c \cos(2\pi f_c t + \Theta)$ with Θ uniform over a full 2π interval. Averaging over phase yields

$$\mu_X(t) = 0, \qquad R_X(\tau) = \frac{A_c^2}{2}\cos(2\pi f_c \tau).$$

The process is WSS; indeed a time shift only rotates a uniformly distributed phase. Its PSD is

$$S_X(f) = \frac{A_c^2}{4}[\delta(f - f_c) + \delta(f + f_c)], \qquad P_X = A_c^2/2.$$

A specified exact phase has prior probability zero for a continuous phase distribution. A noiseless amplitude measurement usually leaves two compatible phases. The two inverse branches can have equal conditional weights under a uniform prior, but that is a statement about a conditional distribution, not a prior probability of 1/2 for a selected phase.

Worked example 7.2 (A restricted phase distribution; Sheet 8, Problem 4.19)**.**
If instead $\Theta \sim \mathrm{Unif}[-\pi/2, \pi/2]$, direct integration gives

$$\mu_X(t) = \frac{2A_c}{\pi}\cos(2\pi f_c t), \qquad R_X(t_1, t_2) = \frac{A_c^2}{2}\cos[2\pi f_c(t_1 - t_2)].$$

The correlation depends only on lag, but the mean is not constant. Thus the process is not WSS. Its statistics are periodic in absolute time, making a cyclostationary description appropriate.

7.3 Gaussian probabilities and transformations

For $Z \sim \mathcal{N}(0,1)$, define

$$\mathrm{Q}(x) = \Pr(Z > x) = \frac{1}{\sqrt{2\pi}}\int_x^\infty e^{-u^2/2}\,\mathrm{d}u = \frac{1}{2}\operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right). \tag{7.2}$$

For $X \sim \mathcal{N}(\mu, \sigma^2)$, standardization gives $\Pr(X > a) = \mathrm{Q}((a-\mu)/\sigma)$. This substitution is the basic step in most Gaussian error calculations.

Worked example 7.3 (A limiter; Sheet 8, Problem 4.13)**.**
Clip $X \sim \mathcal{N}(0, \sigma^2)$ to $[-a, a]$. Within the interval, the output equals the input. Every input above a is mapped to the single point a, and likewise below $-a$. The output law is therefore

$$p_Y(y) = \frac{e^{-y^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}\mathbf{1}_{|y|<a} + \mathrm{Q}(a/\sigma)[\delta(y-a) + \delta(y+a)].$$

Integrating the continuous part gives $1 - 2\,\mathrm{Q}(a/\sigma)$; adding the endpoint masses gives one. Omitting those masses would produce an unnormalized distribution.

7.4 White Gaussian noise through a filter

AWGN combines three assumptions: noise adds to the signal, has flat spectral density over the relevant band, and has jointly Gaussian statistics. The ideal real white-noise model is

$$S_w(f) = \frac{N_0}{2}, \qquad R_w(\tau) = \frac{N_0}{2}\delta(\tau). \tag{7.3}$$

Ideal white noise over all frequencies has infinite power and is interpreted as a generalized process. A receiver filter makes the measured noise power finite:

$$P_n = \frac{N_0}{2}\int_{-\infty}^{\infty} |H(f)|^2\,\mathrm{d}f. \tag{7.4}$$

For an ideal low-pass filter passing $[-B, B]$ with unit gain, $P_n = N_0 B$. For a real bandpass filter with two lobes of width B_T, $P_n = N_0 B_T$. The latter B_T is the width of one RF band; both positive and negative frequencies enter the integral.

For a low-pass filter referenced to DC gain, the one-sided equivalent noise bandwidth is

$$B_{\mathrm{eq}} = \frac{\int_0^\infty |H(f)|^2\,\mathrm{d}f}{|H(0)|^2}.$$

Thus $P_n = N_0|H(0)|^2 B_{\mathrm{eq}}$. A 3-dB bandwidth need not equal the noise bandwidth.

7.5 Thermal noise and noise figure

The available thermal-noise power delivered by a matched source at temperature T into bandwidth B is k_BTB. In this convention, $N_0 = k_BT$, where $k_B \simeq 1.380649 \times 10^{-23}$ J/K. At the reference temperature $T_0 = 290$ K, the available noise density is about -174 dBm/Hz.

A device's noise factor is

$$F = \frac{\mathrm{SNR}_{\mathrm{in}}}{\mathrm{SNR}_{\mathrm{out}}}, \qquad \mathrm{NF} = 10 \log_{10} F.$$

For the conventional reference-temperature model, its equivalent input noise temperature is $T_e = (F-1)T_0$. For cascaded stages with available power gains G_i and noise factors F_i,

$$F_{\mathrm{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots .$$

The first stage matters strongly because its added noise is not suppressed by preceding gain. These formulas assume consistent matched-network and reference-temperature conventions.

7.6 Bandpass noise and quadrature normalization

Suppose a unit-gain bandpass filter admits AWGN over $[f_c - W, f_c + W]$ and its negative-frequency mirror, so $B_T = 2W$. Write

$$n(t) = n_I(t)\cos(2\pi f_c t) - n_Q(t)\sin(2\pi f_c t). \tag{7.5}$$

Multiplication by $2\cos$ or $-2\sin$ and low-pass filtering to $[-W, W]$ gives

$$S_{n_I}(f) = S_{n_Q}(f) = N_0 \mathbf{1}_{|f| \le W}. \tag{7.6}$$

Two RF noise images translate into each baseband output. Each quadrature power is $2N_0W = N_0B_T$. The real bandpass power is

$$P_n = \frac{P_{n_I} + P_{n_Q}}{2} = N_0 B_T, \tag{7.7}$$

not their unweighted sum. For the symmetric flat bandpass spectrum, the baseband cross-spectrum is zero. Since the projected processes are jointly Gaussian, this implies their independence.

Alternatively, using $\sqrt{2}$ in the real-passband representation rescales each quadrature PSD by a factor of two. Either convention works; the waveform normalization and the PSD must change together. Equation (7.6) is the convention used in the next chapter.

Check your understanding

1. Give a stationary process that is not mean ergodic.
2. Find the noise power in a unit-gain ideal baseband filter of bandwidth 10 kHz if $N_0 = 4 \times 10^{-12}$ W/Hz.
3. When does uncorrelated imply independent?

Answers: a nondegenerate random constant; 4×10^{-8} W; for jointly Gaussian variables (or processes when all relevant cross-covariances vanish).

Chapter 8

Performance of analog communication systems

Course connection: Week 5; Sheet 9..

8.1 Defining the SNR before comparing systems

Let P_R denote the received signal power before demodulation, and let the message bandwidth be B_m. Define

$$\gamma = \frac{P_R}{N_0 B_m}, \qquad \mathrm{SNR}_R = \frac{P_R}{N_0 B_T} = \gamma \frac{B_m}{B_T}. \tag{8.1}$$

γ is a common reference for comparing modulation schemes at the same received power, message bandwidth and noise density. SNR_R is the actual pre-detector SNR for an ideal bandpass receiver of width B_T. A nonlinear detector's threshold is imposed on its actual input SNR, not on γ.

8.2 Baseband transmission and equalization

For undistorted baseband transmission followed by a unit-gain ideal low-pass filter, the output SNR is $P_R/(N_0 B_m) = \gamma$.

Suppose instead a channel $H_c(f)$ distorts the signal and noise is added *after* the channel. A zero-forcing equalizer $H_{\mathrm{eq}} = 1/H_c$ over the message band restores the signal but also shapes the noise:

$$\widehat{m} = m + n_{\mathrm{eq}}, \qquad S_{n_{\mathrm{eq}}}(f) = \frac{N_0}{2} |H_{\mathrm{eq}}(f)|^2. \tag{8.2}$$

Near a channel null, the inverse gain becomes large and so does the noise. Equalization cannot recover a lost component at an exact spectral zero.

Worked example 8.1 (Noise enhancement; Sheet 9, Problem 5.2)**.**
Let $S_m(f) = (M_0/2)\,\mathrm{rect}(f/(2B_m))$ and $H_c(f) = (1 + jf/B_m)^{-1}$. Signal power is $M_0 B_m$. The equalizer gives

$$P_n = \frac{N_0}{2} \int_{-B_m}^{B_m} \left(1 + \frac{f^2}{B_m^2}\right) \mathrm{d}f = \frac{4}{3} N_0 B_m.$$

Thus $\mathrm{SNR}_D = 3M_0/(4N_0)$ rather than M_0/N_0. Removing distortion costs $10 \log_{10}(3/4) = -1.249$ dB of SNR in this model.

8.3 AM with a coherent receiver

For received signal $A_R[m_{\mathrm{dc}} + m(t)]\cos(2\pi f_c t)$, coherent multiplication by $2\cos$ followed by message-band filtering gives $A_R m(t) + n_I(t)$ after DC removal. Section 7.6 gives output noise power $2N_0B_m$. Therefore

$$\mathrm{SNR}_D = \frac{A_R^2 P_m}{2N_0 B_m} = \frac{P_m}{m_{\mathrm{dc}}^2 + P_m}\frac{P_R}{N_0 B_m} = \eta_{\mathrm{AM}}\gamma. \tag{8.3}$$

For DSB-SC, $\eta_{\mathrm{AM}} = 1$, so $\mathrm{SNR}_D = \gamma$. Ordinary transmitted-carrier AM spends some received power on the carrier and therefore gives a smaller output SNR for the same P_R.

8.4 Envelope detection and threshold

Let $a(t) = A_R[m_{\mathrm{dc}} + m(t)] > 0$. The noisy envelope is

$$\sqrt{[a(t) + n_I(t)]^2 + n_Q^2(t)}.$$

For small noise relative to the local envelope,

$$\sqrt{[a + n_I]^2 + n_Q^2} \simeq a + n_I + \frac{n_Q^2}{2a}.$$

At first order, the output matches the coherent receiver, so Eq. (8.3) applies. If the envelope approaches zero or input SNR is too low, this expansion fails and distortion becomes important. A quoted threshold such as 6 dB is an exercise-specific approximation, not a universal property of all envelope receivers.

A high average RF SNR does not by itself cure overmodulation. The nonnegative-envelope condition is a separate requirement.

Worked example 8.2 (An AM power budget; Sheet 9, Problem 5.6)**.**
Use $P_m = 0.5$, $B_m = 10$ kHz, $N_0 = 4\times10^{-12}$ W/Hz, channel power loss 70 dB and target $\mathrm{SNR}_D = 10$. With transmitted power measured at the sheet's point T, $P_R = P_T 10^{-7}$. Equation (8.3) gives

$$P_T = \frac{10N_0B_m}{\eta_{\mathrm{AM}}}10^7.$$

For $m_{\mathrm{dc}} = 0.5$, $\eta = 2/3$ and $P_T = 6$ W. For $m_{\mathrm{dc}} = 1$, $\eta = 1/3$ and $P_T = 12$ W. These are the coherent-detection results.

With $B_T = 2B_m$, the respective input SNRs are 7.5 (8.75 dB) and 15 (11.76 dB), above the stated 6 dB threshold. Envelope reception still requires $m(t) + m_{\mathrm{dc}} \geq 0$, which the stated message power alone does not guarantee. In particular, a unit-amplitude sinusoid of power 0.5 overmodulates the $m_{\mathrm{dc}} = 0.5$ case. These qualifications must be checked before assigning the same numbers to an envelope receiver.

8.5 FM noise above threshold

At high input SNR, small quadrature noise perturbs the phase approximately by $n_Q(t)/A_R$. An ideal FM demodulator differentiates the phase and divides by $2\pi k_f$. Its message-referred noise PSD is therefore

$$S_{n_D}(f) \simeq \frac{N_0 f^2}{A_R^2 k_f^2}, \qquad |f| \leq B_m. \tag{8.4}$$

The differentiation cancels the 2π in the discriminator normalization. Integration gives

$$P_{n_D} = \frac{2N_0B_m^3}{3A_R^2k_f^2}, \qquad \mathrm{SNR}_D \simeq 3P_m\left(\frac{k_f}{B_m}\right)^2\gamma, \tag{8.5}$$

where $P_R = A_R^2/2$. With normalized peak message $m_p = 1$, $k_f = \Delta f$. Generally $k_f = \Delta f/m_p$, so the formula becomes $3(P_m/m_p^2)(\Delta f/B_m)^2\gamma$.

Greater deviation can improve output SNR, but also broadens the receiver bandwidth and lowers SNR_R at fixed received power. Below threshold, phase slips and large errors invalidate the small-noise model. Pre-emphasis boosts high message frequencies before modulation, and complementary de-emphasis attenuates them after demodulation, reducing the rising high-frequency noise contribution. The allowable deviation and actual message spectrum constrain this improvement.

Worked example 8.3 (FM threshold changes the design; Sheet 9, Problem 5.8)**.**
Use $B_m = 10$ kHz, $P_m = 0.5$, $m_p = 1$, $N_0 = 4\times10^{-12}$ W/Hz, 70 dB loss, target output SNR 10^4, and the sheet's 6 dB input threshold. Write $\beta_D = \Delta f/B_m$ (a deviation ratio; for a band-edge tone it equals the FM index).

The output target requires $\gamma \geq 10^4/(1.5\beta_D^2)$. Carson bandwidth and the threshold additionally require

$$\gamma \geq 10^{6/10}\frac{B_T}{B_m} \simeq 10^{6/10}\,2(\beta_D+1).$$

Both constraints must hold. Since $N_0B_m10^7 = 0.4$ W,

$$P_T \geq 0.4\max\left\{\frac{10^4}{1.5\beta_D^2},\ 2(\beta_D+1)10^{0.6}\right\}\ \text{W}.$$

β_D	B_T	γ from output	γ from threshold	P_T
10	220 kHz	66.67	87.58	35.03 W
20	420 kHz	16.67	167.21	66.88 W

These values are threshold-based design estimates using Carson bandwidth and the stated simplified threshold. Real receivers need margin. Ignoring threshold gives 26.67 W and 6.67 W, but their actual input SNRs are only 4.81 dB and −4.01 dB. Both fail the exercise's threshold requirement. Increasing deviation is therefore not an unconditional way to reduce transmitter power.

8.6 A comparison with fixed conventions

System	Output SNR	Conditions
Ideal baseband	γ	Message preserved by channel/filter
Coherent DSB-SC	γ	Carrier synchronization
Coherent AM	$\eta_{\mathrm{AM}}\gamma$	Carrier synchronization
Envelope AM	$\simeq \eta_{\mathrm{AM}}\gamma$	No envelope folding; above threshold
FM	$\simeq 3P_m(k_f/B_m)^2\gamma$	Above threshold; adequate RF bandwidth

Check your understanding

1. Why does a zero-forcing equalizer amplify noise near an attenuated band edge?
2. If $B_T = 20B_m$, what is the relation between γ and receiver input SNR?
3. Which two separate conditions must be checked before using the linear envelope-detector SNR?

Answers: it applies inverse channel gain to the added noise; $\mathrm{SNR}_R = \gamma/20$; no envelope folding and sufficiently strong signal relative to noise.

Chapter 9

Matched filters and binary decisions

Course connection: Week 6; Sheet 10 and Sheet 11, Problem 7.4..

9.1 The receiver's sufficient statistic

For a known pulse in AWGN, the receiver need not retain every point of the observed waveform. It can project the observation onto the relevant pulse shape. This concentrates the useful signal into a scalar while rejecting noise components orthogonal to that shape.

Let $r(t) = s(t) + w(t)$, with real white-noise PSD $N_0/2$. A linear filter produces, at decision time T,

$$y_s(T) = \int s(v)h(T-v)\,\mathrm{d}v, \qquad \sigma_n^2 = \frac{N_0}{2}\int |h(v)|^2\,\mathrm{d}v.$$

Cauchy–Schwarz bounds the signal term:

$$|y_s(T)|^2 \le \left(\int |s(v)|^2\,\mathrm{d}v\right)\left(\int |h(T-v)|^2\,\mathrm{d}v\right).$$

Consequently,

$$\frac{|y_s(T)|^2}{\sigma_n^2} \le \frac{2E_s}{N_0}. \tag{9.1}$$

Equality holds for the matched filter

$$h(t) = Ks^*(T-t), \qquad H(f) = KS^*(f)e^{-j2\pi fT}. \tag{9.2}$$

It reverses the pulse in time, conjugates it if needed, and delays it to the chosen decision instant. The arbitrary nonzero gain K scales signal and noise together and does not change SNR.

Key idea. A matched filter maximizes sample SNR for a known waveform in white noise. To minimize binary decision error, match the relevant difference between the candidate waveforms; for antipodal pulses this is the same shape.

If the noise is colored, the optimum frequency weighting instead contains $1/S_w(f)$. One may think of this as whitening the noise before matching. If the channel distorts the pulse, the received pulse is the relevant shape. Matching alone does not guarantee zero ISI in a pulse train.

9.2 Correlation and the output pulse

For a real pulse $p(t)$, $h(t) = p(T_s - t)$ gives

$$(p * h)(t) = R_p(t - T_s).$$

The peak at T_s is E_p. The filter computes the correlation between the observation and the candidate pulse at that instant. A correlator and a matched filter can therefore implement the same decision statistic.

Worked example 9.1 (Rectangular pulses; Sheet 10, Problem 6.15)**.**
For $p(t) = 1$ on $[0, T_s]$ and zero elsewhere, the matched filter has the same rectangular shape. Its output to one unit pulse is

$$y(t) = T_s \,\mathrm{tri}\left(\frac{t - T_s}{T_s}\right).$$

The sequence $\{A, -A, A, A\}$ produces samples $\{AT_s, -AT_s, AT_s, AT_s\}$ at $T_s, 2T_s, 3T_s, 4T_s$. The neighboring triangles overlap between samples but vanish at each other's peaks. The pulse energy is $E_p = T_s$, the bit energy is $E_b = A^2 T_s$, and peak SNR is $2A^2 T_s/N_0$.

Worked example 9.2 (Manchester pulses; Sheet 10, Problem 6.16)**.**
Let $p_M = +1$ on $[0, T_s/2)$ and -1 on $[T_s/2, T_s)$. Time reversal changes its sign, so $h(t) = p_M(T_s - t)$ is not simply the original waveform. Its autocorrelation is

$$R_M(\tau) = \begin{cases} T_s - 3|\tau|, & |\tau| \le T_s/2, \\ |\tau| - T_s, & T_s/2 < |\tau| \le T_s, \\ 0, & |\tau| > T_s. \end{cases}$$

It has $R_M(0) = T_s$, $R_M(T_s/2) = -T_s/2$, and $R_M(T_s) = 0$. Sampling the matched-filter output at bit endpoints again yields $a_n T_s$, and peak SNR is the same as polar NRZ at equal bit energy. The spectra and timing behavior are different even though optimum AWGN performance is identical.

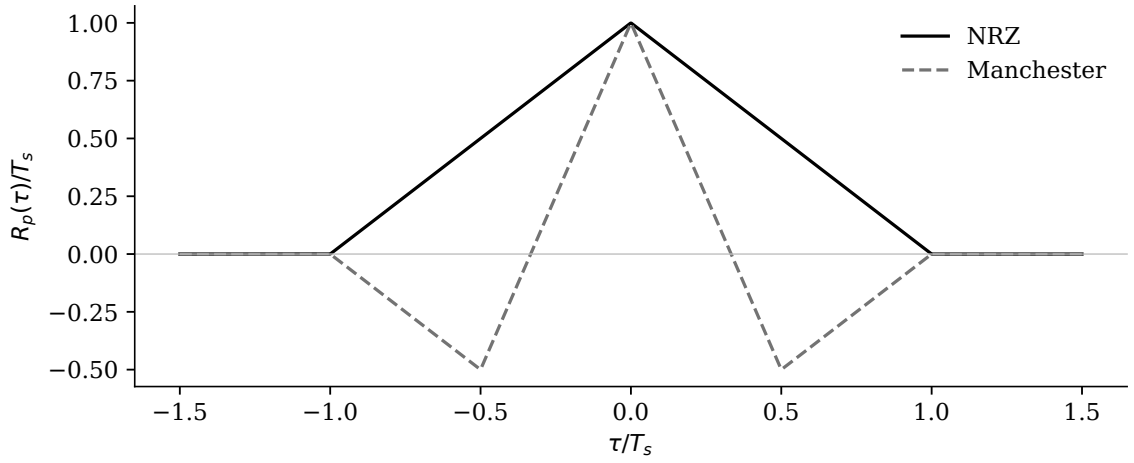


Figure 9.1: Autocorrelations of NRZ and Manchester unit pulses. The matched-filter outputs are delayed copies of these curves. Both have the same peak and zero values at nonzero integer symbol delays.

Worked example 9.3 (Sinc signaling; Sheet 10, Problem 6.18)**.**
Let $p(t) = \mathrm{sinc}(t/T_b)$. Its spectrum is $T_b \,\mathrm{rect}(fT_b)$, so a channel of bandwidth B passes it undistorted only if $B \ge 1/(2T_b)$, giving $R_b \le 2B$.

With matched filter $p(-t)$, $|P(f)|^2 = T_b^2 \,\mathrm{rect}(fT_b)$ gives output pulse $T_b \,\mathrm{sinc}(t/T_b)$. At $t = kT_b$, the desired sample is $a_k T_b$ with no ISI, and noise variance is $N_0 T_b/2$. For $a_k = \pm 1$, $E_b = T_b$ and $P_b = \mathrm{Q}(\sqrt{2E_b/N_0})$. The ideal sinc filter is noncausal; practical versions use delay and truncation.

9.3 Binary Gaussian decisions

Suppose the scalar observation has conditional distributions

$$Y|0 \sim \mathcal{N}(\mu_0, \sigma^2), \qquad Y|1 \sim \mathcal{N}(\mu_1, \sigma^2), \qquad \mu_1 > \mu_0.$$

For threshold λ, decide 1 if $Y > \lambda$. The bit error probability is

$$P_b = p_0 \, \mathrm{Q}\left(\frac{\lambda - \mu_0}{\sigma}\right) + p_1 \, \mathrm{Q}\left(\frac{\mu_1 - \lambda}{\sigma}\right). \tag{9.3}$$

With equal error costs, the minimum-error MAP decision compares $p_1 p(y|1)$ with $p_0 p(y|0)$. Taking logarithms and expanding the squares gives

$$\lambda_{\mathrm{MAP}} = \frac{\mu_0 + \mu_1}{2} + \frac{\sigma^2}{\mu_1 - \mu_0} \ln \frac{p_0}{p_1}. \tag{9.4}$$

Equal priors give the midpoint. If symbol 1 is more likely, the threshold decreases and its decision region expands. Unequal variances generally lead to a quadratic comparison and may require more than one threshold.

9.4 Antipodal signaling and integrate-and-dump

For $s_1(t) = Ap(t)$ and $s_0(t) = -Ap(t)$, correlation with p gives means $\pm AE_p$ and variance $N_0 E_p/2$. Equal priors give threshold zero and

$$P_b = \mathrm{Q}\left(\sqrt{\frac{2E_b}{N_0}}\right), \qquad E_b = A^2 E_p. \tag{9.5}$$

For a rectangular pulse, the correlator is an integrator reset after each bit: $Y = \int_0^{T_b} r(t)\,\mathrm{d}t$. Its means are $\pm AT_b$ and variance $N_0 T_b/2$. Longer integration increases the useful mean and the noise variance, but improves their squared-mean-to-variance ratio in proportion to T_b at fixed amplitude.

9.5 Distance determines binary error probability

For any two equiprobable deterministic signals in AWGN, let

$$d^2 = \int |s_1(t) - s_0(t)|^2 \,\mathrm{d}t.$$

Projecting onto the difference direction reduces the problem to two scalar points separated by d. Their midpoint is $d/2$ from either point, and noise standard deviation is $\sqrt{N_0/2}$. Hence

$$P_b = \mathrm{Q}\left(\frac{d}{\sqrt{2N_0}}\right). \tag{9.6}$$

Antipodal equal-energy signals have $d^2 = 4E_b$. Orthogonal equal-energy binary signals have $d^2 = 2E_b$. For on-off signaling, if the on-symbol energy is E_{on}, the average bit energy is $E_b = E_{\mathrm{on}}/2$ and $d^2 = E_{\mathrm{on}} = 2E_b$.

Worked example 9.4 (Orthogonal and on-off signals; Sheet 11, Problem 7.4)**.**
Orthogonal signals $\sqrt{E}\phi_1$ and $\sqrt{E}\phi_2$ have coordinates $(\sqrt{E}, 0)$ and $(0, \sqrt{E})$, so the decision boundary is $y_1 = y_2$. On-off signals $\sqrt{E}\phi_1$ and zero have threshold $y_1 = \sqrt{E}/2$.

Both give $P_b = \mathrm{Q}(\sqrt{E_b/N_0})$ when expressed in *their own average bit energy.* For $P_b = 10^{-6}$, $\mathrm{Q}^{-1}(10^{-6}) \simeq 4.7534$, so $E_b/N_0 \simeq 22.60 = 13.54$ dB. Antipodal signaling needs half as much energy, 10.53 dB.

Rotation does not change distances or energies. Translation preserves distances and hence error probability for the translated constellation with an adjusted receiver, but generally changes average energy. An error-probability comparison and an energy-efficiency comparison must therefore be kept separate.

Check your understanding

1. Why does matched-filter gain not affect peak SNR?
2. What happens to the MAP threshold when the prior probability of symbol 1 increases?
3. Can changing a pulse from NRZ to Manchester improve ideal antipodal AWGN BER at fixed E_b/N_0?

Answers: squared signal and noise variance scale equally; it decreases; no, although bandwidth and synchronization properties change.

Chapter 10

Signal space and digital carrier modulation

Course connection: Both Week 6 decks; Sheets 11 and 13..

10.1 Waveforms as vectors

For real signals of duration T_s, define $\langle x, y\rangle = \int_0^{T_s} x(t)y(t)\,\mathrm{d}t$. Choose an orthonormal basis $\{\phi_1, \ldots, \phi_N\}$ for the span of the M candidate signals. Then

$$s_m(t) = \sum_{i=1}^{N} s_{mi}\phi_i(t), \qquad s_{mi} = \langle s_m, \phi_i\rangle. \tag{10.1}$$

Orthogonality makes geometry preserve waveform quantities:

$$E_m = \|\mathbf{s}_m\|^2, \qquad d_{m\ell}^2 = \|\mathbf{s}_m - \mathbf{s}_\ell\|^2.$$

Energy is proportional to the *squared* distance from the origin. The number of symbols M differs from the dimension N; always $N \le M$, and the inequality need not be strict.

10.2 Gram–Schmidt construction

Start with $\phi_1 = s_1/\|s_1\|$ for a nonzero first signal. For each subsequent candidate, subtract its projections onto the existing basis:

$$v_m = s_m - \sum_{i=1}^{N_{\text{current}}} \langle s_m, \phi_i\rangle\phi_i. \tag{10.2}$$

If $v_m \neq 0$, normalize it and add it to the basis. If it is zero, the signal already lies in the existing span. Never normalize a zero residual.

Worked example 10.1 (Piecewise-constant signals; Sheet 11, Problem 7.1)**.**
In the sheet, s_1 equals 2 on $[0,1)$, s_2 equals -4 on $[0,2)$, and s_3 equals 3 on $[0,3)$, with each zero elsewhere. A convenient orthonormal basis is

$$\phi_1 = \mathbf{1}_{[0,1)}, \quad \phi_2 = \mathbf{1}_{[1,2)}, \quad \phi_3 = \mathbf{1}_{[2,3)}.$$

All intervals have unit duration. The coordinates are $(2,0,0)$, $(-4,-4,0)$ and $(3,3,3)$, and energies are 4, 32 and 27. For intervals of duration τ rather than one, each basis pulse needs height $1/\sqrt{\tau}$.

10.3 Noise vectors and optimum decisions

Projection onto an orthonormal basis gives

$$\mathbf{y} = \mathbf{s}_m + \mathbf{n}, \qquad \mathbf{n} \sim \mathcal{N}(\mathbf{0}, (N_0/2)I_N). \tag{10.3}$$

To see the variance, use $R_w(t-v) = (N_0/2)\delta(t-v)$ inside the double integral for $\mathbb{E}[n_i n_j]$, obtaining $(N_0/2)\langle\phi_i, \phi_j\rangle = (N_0/2)\delta_{ij}$. Gaussianity then makes the components independent.

The likelihood is

$$p(\mathbf{y}|m) = (\pi N_0)^{-N/2} \exp\left[-\frac{\|\mathbf{y} - \mathbf{s}_m\|^2}{N_0}\right].$$

MAP detection chooses

$$\widehat{m} = \arg\min_m \left\{\|\mathbf{y} - \mathbf{s}_m\|^2 - N_0 \ln p_m\right\}. \tag{10.4}$$

For equal priors this is nearest-neighbor, or maximum-likelihood (ML), detection. The boundaries are perpendicular bisectors. Unequal priors shift them; colored noise changes Euclidean distance to a covariance-weighted distance.

10.4 ASK, BPSK and QPSK

A convenient passband basis, for an integer number of carrier cycles in T_s, is

$$\phi_I(t) = \sqrt{2/T_s}\cos(2\pi f_c t), \qquad \phi_Q(t) = -\sqrt{2/T_s}\sin(2\pi f_c t), \qquad 0 \le t < T_s.$$

The signs match Eq. (5.8). Binary on-off ASK uses points 0 and $\sqrt{E_{\text{on}}}$ on one axis. BPSK uses $\pm\sqrt{E_b}$ and has the BER in Eq. (9.5). Its envelope does not reveal the bit, so coherent or differential phase-sensitive reception is needed.

QPSK uses four points on a circle. With phases $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$, the coordinates are $(\pm\sqrt{E_s/2}, \pm\sqrt{E_s/2})$. Two independently detected sign bits encode each symbol. With Gray labeling,

$$P_b = \mathrm{Q}\left(\sqrt{E_s/N_0}\right) = \mathrm{Q}\left(\sqrt{2E_b/N_0}\right), \qquad P_s = 2P_b - P_b^2. \tag{10.5}$$

The SER follows from $1 - P_s = (1 - P_b)^2$: correct symbol detection requires both independent sign decisions to be correct. QPSK has the same ideal BER as BPSK at equal E_b/N_0, while requiring half the symbol rate for the same bit rate.

10.5 Carrier bandwidth and symbol rate

For linear passband modulation with a raised-cosine overall response,

$$B_T = 2B_{\text{BB}} = (1+\alpha)R_s = \frac{(1+\alpha)R_b}{\log_2 M}. \tag{10.6}$$

For unshaped rectangular symbols, the approximate RF main-lobe width is instead $2R_s$. Rectangular pulses have infinite spectral support; their first-null estimate is not interchangeable with the strict support of an ideal raised-cosine spectrum.

Worked example 10.2 (4-ASK spectrum; Sheet 6, Problem 6.26)**.**
With equiprobable levels $\{-3,-1,1,3\}$, $\mathbb{E}[a^2] = 5$. NRZ baseband PSD is $5T_s\,\mathrm{sinc}^2(fT_s)$. Multiplication by $A_c\cos(2\pi f_c t)$ gives the time-averaged passband PSD

$$S_s(f) = \frac{5A_c^2T_s}{4}\left[\mathrm{sinc}^2((f-f_c)T_s) + \mathrm{sinc}^2((f+f_c)T_s)\right].$$

Total power is $5A_c^2/2$. At $R_b = 1$ Mbit/s, $R_s = 0.5$ Mbaud, so the rectangular-pulse RF main-lobe width is approximately 1 MHz. With raised-cosine roll-off $\alpha = 0.25$, the occupied RF bandwidth would instead be 0.625 MHz.

10.6 Frequency-shift keying

Binary FSK assigns each bit a different tone over T_b. If the waveforms are orthogonal, their distance squared is $2E_b$. Coherent orthogonal BFSK therefore has $P_b = \mathrm{Q}(\sqrt{E_b/N_0})$.

Orthogonality must be checked by integrating the product of the actual time-windowed tones. The minimum spacing depends on phase convention and whether detection is coherent. Spacing $1/T_b$, as in the Sunde-type example in the slides, provides orthogonality under the usual carrier-cycle assumptions and is suitable for phase-insensitive tone separation. A coherent construction with aligned phases can use $1/(2T_b)$ with compatible frequencies. For noncoherent orthogonal BFSK with independent AWGN and equal priors, the standard energy-detector result is $P_b = \frac{1}{2}e^{-E_b/(2N_0)}$; it describes a different receiver from coherent correlation.

10.7 M-PSK and square QAM

For M-PSK, equally spaced phases on a radius-$\sqrt{E_s}$ circle give

$$d_{\min}^2 = 4E_s\sin^2(\pi/M). \tag{10.7}$$

Increasing M carries more bits per symbol, but reduces spacing at fixed energy. At sufficiently high SNR, for $M \geq 4$,

$$P_s \simeq 2\,\mathrm{Q}\left(\sqrt{2E_s/N_0}\sin(\pi/M)\right). \tag{10.8}$$

This nearest-neighbor approximation is not the exact SER for arbitrary M or SNR.

Square M-QAM, with $M = L^2$ and even L, uses L equally spaced levels on each axis. If adjacent points are distance d apart,

$$E_s = \frac{(M-1)d^2}{6}, \quad q = \mathrm{Q}\left(\sqrt{\frac{3E_s}{(M-1)N_0}}\right), \quad P_s = 1 - \left[1 - 2\left(1 - \frac{1}{\sqrt{M}}\right)q\right]^2. \tag{10.9}$$

This SER is exact for equiprobable square QAM with ideal coherent detection in AWGN. It follows by treating each axis as independent PAM. Interior PAM levels have two error tails; edge levels have one.

Worked example 10.3 (An asymmetric four-point constellation; Sheet 11, Problem 7.3)**.**
The signals $A\cos\omega_c t$, $A\cos(\omega_c t + \pi/2)$ and their doubled-amplitude versions span only two dimensions. In the basis above, the coordinates are $(a,0),(0,a),(2a,0),(0,2a)$, where $a = A\sqrt{T_s/2}$.

Thus $d_{\min} = a$, $d_{\max} = 2\sqrt{2}a = 2A\sqrt{T_s}$, and $\overline{E}_s = (1+1+4+4)a^2/4 = 5A^2T_s/4$. An opposite sine-basis sign reflects the constellation without changing these quantities.

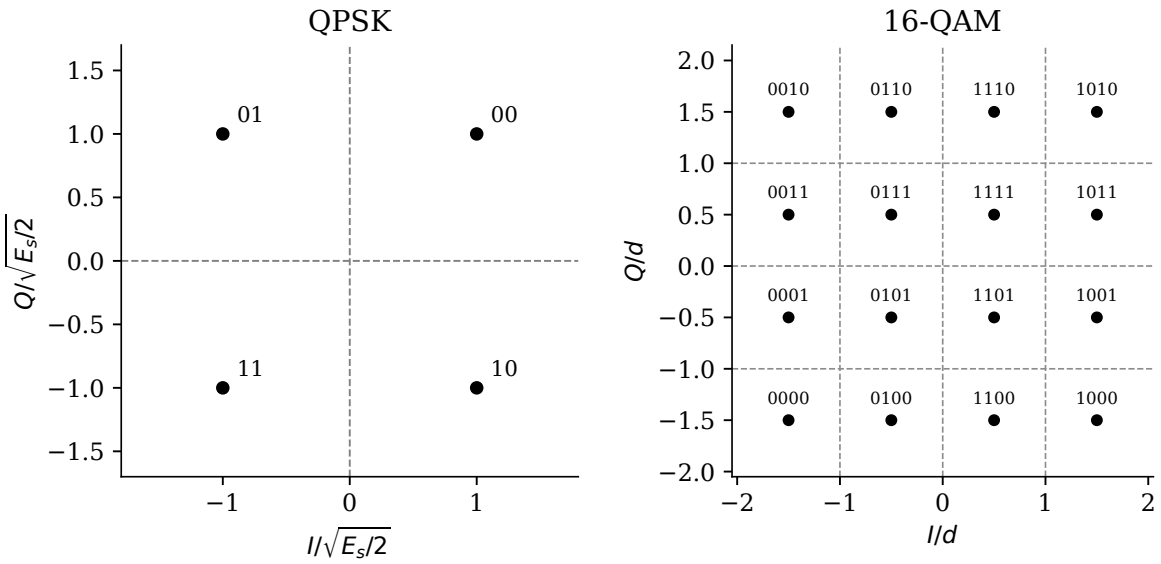


Figure 10.1: QPSK and 16-QAM with representative Gray labels and ML decision boundaries. Axis labels are energy-normalized signal coordinates. QPSK sign decisions are independent; 16-QAM also requires amplitude decisions.

Worked example 10.4 (Exact QPSK SER; Sheet 13, Problems 7.5 and 7.7)**.**
The waveform amplitude is one, $R_s = 1$ Mbaud and $N_0 = 10^{-7}$ W/Hz. Then $T_s = 10^{-6}$ s and $E_s = T_s/2 = 5 \times 10^{-7}$ J. Thus $E_s/N_0 = 5$ and

$$q = \mathrm{Q}(\sqrt{5}) \simeq 0.012674, \qquad P_s = 2q - q^2 \simeq 0.02519.$$

The first-order expression $2q \simeq 0.02535$ slightly overestimates SER. A 45-degree rotation transforms the axial four-point constellation in the sheet into the usual quadrant constellation without changing its error probability.

Worked example 10.5 (16-QAM regions; Sheet 13, Problem 7.6)**.**
For coordinates $\{\pm d/2, \pm 3d/2\}$ on each axis, the boundaries are $x, y \in \{-d, 0, d\}$. Let $q = \mathrm{Q}(d/\sqrt{2N_0})$. The four interior points have correct probability $(1-2q)^2$, the four corners $(1-q)^2$, and the eight other boundary points $(1-2q)(1-q)$. Averaging gives

$$P_s = 3q - \frac{9}{4}q^2.$$

This is the $M = 16$ case of the square-QAM formula. It includes simultaneous errors on both axes rather than counting them twice.

10.8 Union bounds and the difference between BER and SER

For equiprobable symbols, pairwise comparison against symbol j when i was sent has probability

$$\Pr\{j \text{ has larger likelihood than } i \mid i\} = \mathrm{Q}\left(\frac{d_{ij}}{\sqrt{2N_0}}\right).$$

The union bound gives

$$P_s \le \frac{1}{M} \sum_i \sum_{j \neq i} \mathrm{Q}\left(\frac{d_{ij}}{\sqrt{2N_0}}\right). \tag{10.10}$$

Events overlap, so the bound is generally not exact and can exceed one. Retaining only nearest neighbors usually gives a high-SNR approximation, not automatically an upper bound. For QPSK specifically, the two adjacent decision-boundary crossings already cover the error region, so $P_s \le 2\,\mathrm{Q}(\sqrt{E_s/N_0})$ is a valid tighter bound.

If symbols carry $k = \log_2 M$ bits, one incorrect symbol contains between one and k incorrect bits. Thus

$$\frac{P_s}{k} \le P_b \le P_s. \tag{10.11}$$

More precisely, $P_b = (Mk)^{-1} \sum_i \sum_{j \neq i} d_H(b_i, b_j) \Pr(\widehat{m} = j \mid i)$ for equiprobable symbols. Gray labels make adjacent symbols differ by one bit. Therefore $P_b \simeq P_s/k$ when nearest-neighbor errors dominate. Gray labeling does not itself detect or correct errors; it reduces the bit cost of likely symbol errors.

For a symmetric orthogonal M-symbol detector, wrong decisions are equally likely among the other labels. All k-bit words then give $P_b = MP_s/[2(M-1)]$. This is a special symmetry result, not a generic BER–SER relation.

Check your understanding

1. How many signal-space dimensions does ordinary QPSK require?
2. A 64-QAM link uses $\alpha = 0.2$ at 12 Mbit/s. What RF bandwidth is required?
3. Why does $P_s/\log_2 M$ usually underestimate the exact BER outside the high-SNR Gray-coded regime?

Answers: two; 2.4 MHz; a symbol error may change several bits.

Chapter 11

System operation, information and coding

Course connection: Week 7, with the serial-link examples in Week 6..

11.1 Decibels and link budgets

A power ratio is expressed as $G_{\mathrm{dB}} = 10\log_{10}(P_2/P_1)$. For voltage amplitudes across equal resistances, $G_{\mathrm{dB}} = 20\log_{10}|V_2/V_1|$. If the resistances differ, use the power ratio explicitly. Absolute reference levels are

$$P_{\mathrm{dBm}} = 10\log_{10}\frac{P}{1\ \mathrm{mW}}, \quad P_{\mathrm{dBW}} = 10\log_{10}\frac{P}{1\ \mathrm{W}}, \quad P_{\mathrm{dBm}} = P_{\mathrm{dBW}} + 30. \tag{11.1}$$

Gains multiply in linear units and add in dB. Losses subtract. One cannot add two independent powers by adding their dBm values; convert them to watts, add, and convert back.

For an ideal free-space far-field link with matched polarization and antenna alignment,

$$P_R = P_T G_T G_R \left(\frac{\lambda}{4\pi r}\right)^2. \tag{11.2}$$

In dB units, the free-space path loss is $20\log_{10}(4\pi r/\lambda)$. Real links may also have cable loss, absorption, blockage, multipath and fading. The simple free-space expression does not describe these effects.

Worked example 11.1 (A link calculation)**.**
Suppose $P_T = 20$ dBm, antenna gains are 2 dB each, propagation loss is 80 dB and implementation losses total 3 dB. The received power is $20 + 2 + 2 - 80 - 3 = -59$ dBm.

For 1 MHz noise bandwidth and 5 dB noise figure at the reference thermal temperature, the input-referred noise is approximately $-174 + 60 + 5 = -109$ dBm. The corresponding SNR is 50 dB. This is an illustrative static budget; interference, fading and implementation penalties must be included when applicable.

11.2 Eye diagrams, timing and constellations

An eye diagram overlays successive waveform segments synchronized to the symbol clock. Its vertical opening indicates amplitude margin; its horizontal opening indicates timing margin. ISI splits or spreads the possible trajectories, noise thickens them, and jitter spreads their crossing

times. The best sampling instant is near the largest useful opening, subject to the actual noise and ISI statistics.

For two Gaussian level distributions, a common eye-quality measure is

$$Q_{\text{eye}} = \frac{\mu_1 - \mu_0}{\sigma_1 + \sigma_0}. \tag{11.3}$$

This is an amplitude-to-noise measure, not a power SNR. With equal variances, equal priors and midpoint threshold, $P_b = \mathrm{Q}(Q_{\text{eye}})$. With unequal variances, a threshold equating the two normalized distances also gives this expression for the two tails, but need not be the exact MAP threshold.

A sampled I/Q constellation provides complementary diagnostics. Random circular spreading is consistent with AWGN; common rotation suggests phase offset; changing rotation suggests frequency offset; unequal axis gains or nonorthogonality suggest I/Q imbalance. Such patterns guide diagnosis but do not uniquely identify a physical cause.

11.3 Framing, differential signaling and multiplexing

An asynchronous serial receiver needs to identify symbol timing and word boundaries. Start and stop intervals provide framing; parity can detect some errors. The payload rate is smaller than the wire rate. For example, eight data bits plus one start and one stop bit give a payload fraction 8/10, before any additional checks.

Differential signaling measures the voltage difference between two conductors. A perturbation that couples equally into both appears as common mode and is reduced by a receiver with good common-mode rejection. Real rejection is finite and depends on balance, routing and receiver characteristics. Transition-based codes and bit stuffing constrain long runs to assist clock recovery. These functions are distinct from payload error correction.

Time-division multiplexing allocates different time slots to users. Frequency-division multiplexing allocates different frequency bands, usually with guard bands. Orthogonal frequency-division multiplexing uses subcarriers separated by $1/T_u$ over a useful symbol duration T_u:

$$\int_0^{T_u} e^{j2\pi(k-\ell)t/T_u}\,\mathrm{d}t = 0 \quad (k \neq \ell). \tag{11.4}$$

Their spectra may overlap while the symbol-interval inner products remain zero. A cyclic prefix can turn finite multipath convolution into circular convolution over the useful interval when it covers the channel memory, at the cost of overhead. Timing and frequency errors can destroy orthogonality. Code-division systems separate users by spreading sequences; perfect orthogonality is a design option, not a requirement satisfied by every practical multiuser link.

11.4 Information per message and entropy

The self-information of an event of probability p is $-\log_2 p$ bits. The entropy of a discrete source is its average self-information:

$$H(X) = -\sum_x p(x)\log_2 p(x). \tag{11.5}$$

For a binary source,

$$H_2(p) = -p\log_2 p - (1-p)\log_2(1-p),$$

with $0 \log_2 0 = 0$. Entropy is one bit for equiprobable symbols and zero for a deterministic source. Source coding exploits predictable structure to reduce average description length. Channel coding instead introduces controlled redundancy to combat errors.

A binary symmetric channel flips each bit independently with probability p. Its capacity is $1 - H_2(p)$ bits *per channel use*. At $p = 0.01$ it is approximately 0.9192 bits/use; at $p = 0.5$ it is zero. To obtain bit/s, multiply by uses per second. A use rate is not automatically a bandwidth in hertz.

11.5 AWGN capacity and the energy limit

For the ideal bandlimited AWGN channel under an average-power constraint,

$$C = B \log_2 \left(1 + \frac{P}{N_0 B}\right). \tag{11.6}$$

Here B and $N_0 B$ use consistent physical channel-bandwidth and noise-power conventions. For any fixed rate below capacity, suitable coding with sufficiently long blocks can make error probability arbitrarily small under the model. The formula does not specify the practical code, latency, finite-block performance or receiver implementation.

Writing spectral efficiency $\eta_s = R/B$ and $E_b = P/R$ for the energy per *information* bit, reliable communication requires

$$\frac{E_b}{N_0} \geq \frac{2^{\eta_s} - 1}{\eta_s}. \tag{11.7}$$

Taking $\eta_s \to 0$ gives $E_b/N_0 \geq \ln 2$, or -1.592 dB. This limiting value occurs at vanishing spectral efficiency; it is not an uncoded BPSK operating point. At finite efficiency, the required ratio is larger.

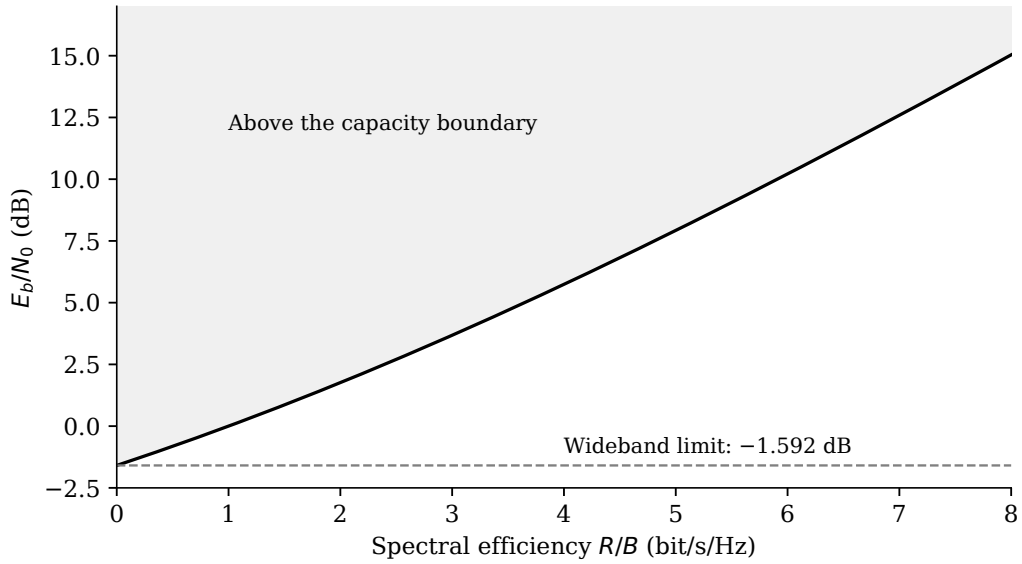


Figure 11.1: The AWGN capacity boundary expressed as energy per information bit versus spectral efficiency. Reliable operation requires an energy at least as large as the boundary value; finite-length practical systems require additional margin.

Worked example 11.2 (Capacity calculation)**.**
An ideal AWGN channel with $B = 3.5$ kHz and SNR 30 dB has

$$C = 3500 \log_2(1001) \simeq 34.89 \text{ kbit/s}.$$

Use the linear SNR 1000 inside the logarithm. A proposed modulation rate does not establish reliable operation: it must satisfy both its waveform bandwidth constraint and its required noise performance.

11.6 Error detection and forward error correction

An (n, k) binary block code maps k information bits into n coded bits. Its rate is $r_c = k/n$. If the payload information rate is R_{info}, the coded bit rate is $R_{\text{coded}} = R_{\text{info}}/r_c$, before other overhead. For fixed-length M-ary modulation,

$$R_s = \frac{R_{\text{info}}}{r_c \log_2 M}, \qquad B_T = \frac{(1+\alpha)R_{\text{info}}}{r_c \log_2 M}.$$

No universal lower bound of 1/2 applies to code rate; for example, the repetition code below has rate 1/3.

The Hamming distance between two binary words counts their differing positions. A code with minimum distance $d_{\min}$ can detect every pattern of up to $d_{\min} - 1$ bit errors, or correct every pattern of up to $\lfloor (d_{\min} - 1)/2 \rfloor$ errors. These are separate guarantees. A parity bit detects every odd number of flips, but cannot generally locate and correct an error.

A systematic code retains the original information bits in specified codeword positions. A convolutional encoder instead has memory: its outputs depend on current and preceding input bits. A simple illustrative rate-1/2 encoder outputs $(u_n, u_n \oplus u_{n-1})$. Its trellis records the memory state. Viterbi decoding is an algorithm for finding a likely trellis path, not a code family by itself.

Hard-decision decoding retains only the decided bit or symbol. Soft decoding retains reliability information such as distances or log-likelihood ratios, which can improve performance. An interleaver redistributes a burst across multiple codewords but does not correct errors by itself. Detection followed by retransmission and forward error correction are different strategies and can be combined.

Worked example 11.3 (The three-bit repetition code)**.**
Encode $0 \mapsto 000$ and $1 \mapsto 111$. Here $d_{\min} = 3$ and majority voting corrects one error. For independent coded-bit errors of probability p,

$$P_{\text{decoded}} = \binom{3}{2} p^2(1-p) + p^3 = 3p^2 - 2p^3. \tag{11.8}$$

At $p = 0.01$, this is 2.98×10^{-4}. The comparison holds at the stated coded-bit error probability. At fixed total energy per information bit, the three transmitted copies have less energy each than a single uncoded bit; the physical error probability must be recalculated. Redundancy is not free energy or free bandwidth.

For a decoder guaranteed to correct up to t errors in an n-bit block, $\sum_{j=t+1}^{n} \binom{n}{j} p^j (1-p)^{n-j}$ is the probability of exceeding that guarantee under independent bit errors. It is an upper bound on block failure if all patterns up to t are corrected; equality requires that all heavier patterns fail. Decoded BER also depends on the mapping and decoder.

11.7 A complete design workflow

Begin with the information requirement. Convert source samples to payload bits, add framing and coding overhead, then choose a constellation and obtain the symbol rate. Next determine

bandwidth from the actual pulse shape. Calculate received power and filtered noise, translate them into the relevant E_b/N_0 or SNR, and check the target error rate. Finally, check timing, synchronization, overload, nonlinear thresholds and implementation margin.

Worked example 11.4 (An end-to-end digital design)**.**
Four analog channels of bandwidth 4 kHz use $f_a = 10$ ksample/s and eight-bit PCM. Their payload rate is $4(10{,}000)(8) = 320$ kbit/s. A rate-1/2 code gives 640 kbit/s before framing. QPSK gives 320 kbaud, and $\alpha = 0.25$ requires 400 kHz RF bandwidth.

A bit energy computed as P/R_{coded} refers to a coded modulation bit, while P/R_{info} refers to an information bit. They differ by $1/r_c = 2$. The uncoded QPSK BER expression predicts channel decisions before decoding; predicting the final decoded BER requires the chosen code and decoder.

Check your understanding

1. Can Gray labeling replace a parity check?
2. Why must capacity calculations specify bits per use versus bits per second?
3. If a rate-1/2 code is added while payload rate and modulation order stay fixed, what happens to symbol rate?

Answers: no; a use has no prescribed duration or bandwidth; symbol rate doubles, before considering other overhead.

Appendix A

Formula reference

Signals, sampling and digitization

$$X(f) = \int x(t) e^{-j2\pi ft}\,\mathrm{d}t, \qquad \operatorname{sinc} u = \frac{\sin \pi u}{\pi u},$$

$$E_x = \int |x(t)|^2\,\mathrm{d}t = \int |X(f)|^2\,\mathrm{d}f, \qquad P_y = \int |H(f)|^2 S_x(f)\,\mathrm{d}f,$$

$$f_a > 2B_m \quad \text{(safe ideal sampling)}, \qquad R_b = \nu f_a \quad \text{(one PCM stream)},$$

$$\Delta = 2A/2^{\nu}, \qquad \sigma_q^2 \simeq \Delta^2/12,$$

$$\mathrm{SQNR} \simeq (3P_m/A^2)2^{2\nu}, \qquad R_b = R_s \log_2 M,$$

$$B_{\mathrm{BB}} = (1+\alpha)R_s/2, \qquad B_T = (1+\alpha)R_s \quad \text{(linear passband)}.$$

The sampling equality $f_a = 2B_m$ is an ideal limit with endpoint qualifications. Rectangular-pulse first-null widths are R_s baseband and $2R_s$ passband; their strict bandwidth is infinite.

Analog modulation and noise

$$P_{\mathrm{AM}} = \tfrac{1}{2}A_c^2(m_{\mathrm{dc}}^2 + P_m), \qquad \eta_{\mathrm{AM}} = \frac{P_m}{m_{\mathrm{dc}}^2 + P_m},$$

$$f_i = (2\pi)^{-1}\theta'(t), \qquad \Delta f_{\mathrm{FM}} = k_f m_p,$$

$$B_{T,\mathrm{FM}} \simeq 2(\Delta f + B_m), \qquad P_n = \frac{N_0}{2}\int |H|^2\,\mathrm{d}f,$$

$$\gamma = \frac{P_R}{N_0 B_m}, \qquad \mathrm{SNR}_R = \frac{P_R}{N_0 B_T},$$

$$\mathrm{SNR}_{D,\mathrm{AM}} = \eta_{\mathrm{AM}}\gamma, \qquad \mathrm{SNR}_{D,\mathrm{FM}} \simeq 3P_m(k_f/B_m)^2\gamma.$$

Envelope-AM and FM output formulas require operation above threshold; envelope AM also requires no folding. Noise quadratures use the normalization in Section 7.6.

Detection and information

$$h_{\mathrm{MF}}(t) = K s^*(T-t), \qquad \mathrm{SNR}_{\mathrm{peak}} = 2E_s/N_0,$$

$$P_{b,\mathrm{binary}} = \mathrm{Q}(d/\sqrt{2N_0}), \qquad P_{b,\mathrm{BPSK}} = \mathrm{Q}(\sqrt{2E_b/N_0}),$$

$$P_{b,\text{coh orth}} = \mathrm{Q}(\sqrt{E_b/N_0}), \qquad P_{s,\mathrm{QPSK}} = 2q - q^2, \quad q = \mathrm{Q}(\sqrt{E_s/N_0}),$$

$$C_{\mathrm{AWGN}} = B\log_2(1 + P/(N_0 B)), \qquad C_{\mathrm{BSC}} = 1 - H_2(p) \quad \text{bits/use},$$

$$E_b/N_0 \geq (2^{R/B} - 1)/(R/B), \qquad r_c = k/n.$$

Binary-distance formulas assume equal priors and coherent AWGN detection. Information-theory E_b is per information bit; uncoded-modulation E_b is per modulation bit. Keep these conventions explicit when adding a code.

Appendix B

Problem-sheet study guide

Use the original exercise numbers below to find the corresponding practice. The first number in a problem identifier belongs to the source exercise collection, not to the chapter numbering of these notes.

Sheet	Problems	Method to practise	Ch.
1	1.8; 1.30; 1.32	Exponential transform; evaluate a filter at spectral lines; piecewise convolution	1–2
2	2.3; 2.9	Correlation by overlap; energy/PSD distinction; spectral energy integration	2
3	6.4; 6.5; 6.6; 6.11	Interpolation droop; sampling and bit budgets; distribution-dependent SQNR	3
4	6.12; 6.13; 6.20; 6.21	Compander inversion; mean and variance in PAM PSD; roll-off and rate	3–4
5	3.3; 3.4; 3.6; 3.10	AM power and spectra; rectifier gain; envelope folding; synchronization	5
6	6.23; 6.25; 6.26	TDM rate budget; number of users; first-null and shaped bandwidths	4, 10
7	3.19; 3.21; 3.24; 3.28	Differentiate phase; PM jumps; FM spectrum; I/Q demodulator bandwidth	6
8	4.13; 4.18; 4.19; 4.29	Gaussian tails; stationarity; phase distributions; I/Q noise normalization	7
9	5.2; 5.6; 5.8	Noise after equalization; received/transmitted power; detector thresholds	8
10	6.15; 6.16; 6.18	Time reversal; autocorrelation peaks; zero ISI; matched-filter BER	9
11	7.1; 7.3; 7.4	Gram–Schmidt; coordinates and energies; binary decision regions	9–10
12	6.24; 6.14	Joint PCM and pulse-shaping design; PAM mean, variance and lines	3–4
13	7.5; 7.6; 7.7	Exact QPSK and QAM errors; decision regions; bounds versus approximations	10

Additional revision problems

R1. A zero-mean analog message has peak 2 V and power 0.4 V^2. It is sampled at 20 ksample/s with eight-bit uniform PCM. Find the step, approximate SQNR and bit rate.

R2. Transmit that PCM stream using 16-QAM with $\alpha = 0.35$, without coding or framing. Find the symbol rate and RF bandwidth.

R3. For $s(t) = A_c[1 + 0.8\cos(2\pi f_m t)]\cos(2\pi f_c t)$, find the fraction of power in the sidebands and the gain needed after an ideal full-wave rectifier and DC removal to recover $0.8\cos(2\pi f_m t)$.

R4. In a coherent binary receiver, $Y|0 \sim \mathcal{N}(-1, 0.25)$ and $Y|1 \sim \mathcal{N}(1, 0.25)$. Find the MAP threshold for priors $p_0 = 0.8, p_1 = 0.2$ and express BER using Q.

R5. A unit-amplitude binary antipodal pulse has duration $2\,\mu$s across a $1\,\Omega$ load. If $N_0 = 10^{-7}$ W/Hz, find E_b/N_0 and BER.

R6. A QPSK link with $R_s = 1$ Mbaud uses a rate-3/4 code and no other overhead. Find the information rate and the relation between energy per information bit and energy per transmitted coded bit.

Revision answers

R1. $\Delta = 4/256 = 0.015625$ V; SQNR $= 0.4/(\Delta^2/12) = 19660.8$, or 42.94 dB; $R_b = 160$ kbit/s.

R2. $R_s = 40$ kbaud and $B_T = 54$ kHz.

R3. $\eta = 0.8^2/(2 + 0.8^2) = 0.2424$; gain $\pi/(2A_c)$.

R4. $\lambda = (0.25/2)\ln 4 = 0.17329$; $P_b = 0.8\,\mathrm{Q}((\lambda + 1)/0.5) + 0.2\,\mathrm{Q}((1 - \lambda)/0.5)$.

R5. $E_b = 2 \times 10^{-6}$ J, $E_b/N_0 = 20$, and $P_b = \mathrm{Q}(\sqrt{40}) \simeq 1.27 \times 10^{-10}$.

R6. $R_{\text{info}} = 1.5$ Mbit/s; $E_{b,\text{info}} = E_{b,\text{coded}}/r_c = (4/3)E_{b,\text{coded}}$ at the same power.

Appendix C

Clarifications relative to the supplied material

The main chapters use a consistent set of conventions and explicitly stated approximations. The following notes explain why some expressions or numerical results differ from the supplied slides and corrected sheets. They are provided so that a student comparing documents can identify the source of a discrepancy.

1. **Sampling and sinc: Sheet 3, Problems 6.4 and 6.6.** Several inequalities are reversed in the solutions. Non-aliasing requires a sample rate at least $2B_m$ in the ideal limit, with practical margin above it. With $\operatorname{sinc} u = \sin(\pi u)/(\pi u)$, linear interpolation uses $\operatorname{sinc}^2(fT_a)$, not $\operatorname{sinc}^2(\pi f T_a)$. The 7-bit, 20 ksample/s rate is 140 kbit/s.

2. **Quantizer terminology: Week 2.** Quantization is part of analog-to-digital conversion, not a DAC operation. Zero mean alone does not imply a distribution concentrated near zero; companding should be justified by the actual amplitude statistics.

3. **Compander spacing: Sheet 4, Problem 6.12.** Distances between reconstructed levels and widths between cell boundaries are different. For the stated midpoint convention the largest level spacing is approximately 1.666 V; the outermost cell width is about 1.702 V.

4. **Unit-energy line-code comparisons: Sheet 4, Problem 6.13.** Setting all pulse amplitudes to one does not give equal average bit energy for polar, unipolar and RZ formats. Chapter 4 states the required normalizations.

5. **Rectifier scaling: Sheet 5, Problem 3.4.** The full-wave rectifier low-frequency gain is $2/\pi$, because both translated spectral copies contribute. The intermediate value A_c/π in that solution misses one copy. The $\pi/5$ gain in Problem 3.6 is consistent with $2/\pi$.

6. **Time-varying phase error: Sheet 5, Problem 3.10.** A term proportional to $\cos\theta(t)$ is not a DC term when θ varies. Its removal requires filtering assumptions. Under overmodulation, a finite low-pass filter also acts on the newly generated harmonics, rather than necessarily retaining the entire folded envelope.

7. **PM discontinuities and FM approximation: Sheet 7, Problems 3.21 and 3.24.** Discontinuous PM phase produces frequency impulses, so a finite maximum over smooth segments is not a global peak. The wideband quasi-stationary spectral approximation also requires large deviation relative to message bandwidth; slow variation relative to the carrier alone is insufficient. A deterministic tone has a line spectrum, even when a continuous envelope is useful.

8. **I/Q FM bandwidth and phase recovery: Sheet 7, Problem 3.28.** The I/Q filters must pass the modulated complex envelope, which is approximately as wide as $\Delta f + B_m$ under Carson's estimate. It can be much wider than the message. Recover phase with quadrant-aware atan2 and unwrapping.

9. **Stationarity and phase probabilities: Week 5 and Sheet 8, Problem 4.18.** WSS does not imply ergodicity. An exact value of a continuously uniform phase has prior probability zero. A two-branch posterior after an ideal amplitude measurement is a different statement, and replacing the prior by a two-point distribution changes the model.

10. **Noise normalization: Week 5 and Sheet 9, Problem 5.6.** With $n = n_I \cos - n_Q \sin$ and original two-sided RF PSD $N_0/2$, baseband quadrature PSD is N_0 after mixing by factors of two. Consequently $P_n = (P_{n_I} + P_{n_Q})/2$. Keeping the same signal normalization but using quadrature PSD $N_0/2$ doubles the predicted SNR incorrectly. Example 8.2 gives coherent transmitter powers 6 W and 12 W, with additional envelope-detector qualifications.

11. **FM threshold and power reference: Sheet 9, Problem 5.8.** The detector threshold applies to $\mathrm{SNR}_R = P_R/(N_0 B_T)$, not to $\gamma = P_R/(N_0 B_m)$. Example 8.3 checks both constraints. If P_T is already measured after the transmitter gain at point T, that gain is not divided out a second time.

12. **PAM spectra and SQNR: Sheet 12.** Independent nonzero-mean amplitudes have nonzero off-diagonal correlation. In the variance-plus-mean decomposition, the exponential comb sum includes all integers. The triangular input gives SQNR $L_q^2/2$, not $3L_q^2/2$. For baseband first-null bandwidth R_s, the nominal fixed-length bit efficiency is $\log_2 M$ bit/s/Hz; $\log_2 M/2$ applies to the corresponding rectangular-pulse passband width $2R_s$.

13. **Constellation geometry and labeling: Week 6.** Signal energy is squared distance from the origin, and an M-symbol constellation has $N \leq M$ signal-space dimensions. Two different signals need not be linearly independent. Rotations preserve energy and distance, while translations preserve distance but generally change energy. 64-QAM and 256-QAM carry six and eight bits per symbol, respectively.

14. **Bounds and Gray coding: Weeks 6–7.** A nearest-neighbor approximation is not automatically a general union bound. Gray labeling minimizes bit changes for adjacent decisions but introduces no error-correcting redundancy.

15. **Operational formulas: Week 7.** The eye expression $(\mu_1 - \mu_0)/(\sigma_0 + \sigma_1)$ is an eye-quality factor, not directly a power SNR. Binary-symmetric-channel capacity is in bits/use. Code rates need not exceed 1/2, and a binomial tail for more than t errors is not automatically an exact decoded BER. Illustrative historical interface and coding examples are not treated as current specifications in these notes.

Appendix D

Source material and further reading

Course sources

Emmanuel Zambrini Cruzeiro, *Telecommunications*, Instituto Superior Técnico, academic year 2025–2026. The supplied archive contains the following primary lecture decks:

1. `Week1-Signals2025.pptx`, 35 slides.
2. `Week2-Digitization2025.pptx`, 32 slides.
3. `Week3-ModAM2025.pptx`, 26 slides.
4. `Week4-ModFM2025.pptx`, 24 slides.
5. `Week5-Noise2025.pptx`, 23 slides.
6. `Week6-DigitalTransmission2025.pptx`, 49 slides, and its supplied AutoRecovered version, 56 slides.
7. `Week7-Operation.pptx`, 25 slides.

The supplementary deck, `week1 - complemento.pptx`, contains 91 slides and credits slides by Simon Haykin and Michael Moher. It develops Fourier analysis and correlation in greater detail. The thirteen `SerieN_corr` PDFs supply the exercise statements and solutions used throughout these notes. Some original solution illustrations credit Francisco Sena; the figures in the present notes are newly drawn from the formulas rather than copied from those illustrations.

Reading identified in the lectures

The primary decks identify Simon Haykin and Michael Moher, *Communication Systems*, fifth edition (student edition), as the course reading. The chapter references below reproduce the reading guidance in those decks; pagination may depend on the edition. The supplementary deck is headed *Introduction to Analog and Digital Communications*, so its chapter numbering should not be assumed interchangeable with the primary-deck references.

Topic	Reading indicated by the primary decks
Signals and systems	Chapter 2; Sections 5.7–5.11
Digitization and baseband transmission	Sections 7.1, 7.8, 8.4 and 8.6
Amplitude modulation	Sections 3.1–3.5
Frequency and phase modulation	Chapter 4
Analog systems with noise	Sections 6.1–6.5
Digital reception and modulation	Section 8.2; Sections 9.1–9.5
Operation and coding	Section 8.6; Chapter 10

The historical foundations cited in the lectures include Harry Nyquist, “Certain Topics in Telegraph Transmission Theory” (1928), and Claude E. Shannon, “A Mathematical Theory of Communication” (1948). These notes use the supplied course materials as their source basis and develop the derivations directly; they do not reproduce textbook passages.